

# An atlas of the aging mouse proteome reveals the features of age-related post-transcriptional dysregulation

Received: 3 July 2024

Accepted: 24 September 2024

Published online: 02 October 2024

 Check for updates

Masaki Takasugi<sup>1,7</sup> , Yoshiki Nonaka<sup>1,6</sup>, Kazuaki Takemura<sup>1,6</sup>, Yuya Yoshida<sup>1,6</sup>, Frank Stein<sup>2</sup>, Jennifer J. Schwarz<sup>2</sup>, Jun Adachi<sup>3</sup>, Junko Satoh<sup>4</sup>, Shinji Ito<sup>4</sup>, Gregory Tomblin<sup>5</sup>, Seyed Ali Biashad<sup>5</sup>, Andrei Seluanov<sup>5</sup>, Vera Gorbunova<sup>5</sup> & Naoko Ohtani<sup>1,7</sup> 

To what extent and how post-transcriptional dysregulation affects aging proteome remains unclear. Here, we provide proteomic data of whole-tissue lysates (WTL) and low-solubility protein-enriched fractions (LSF) of major tissues collected from mice of 6, 15, 24, and 30 months of age. Low-solubility proteins are preferentially affected by age and the analysis of LSF doubles the number of proteins identified to be differentially expressed with age. Simultaneous analysis of proteome and transcriptome using the same tissue homogenates reveals the features of age-related post-transcriptional dysregulation. Post-transcriptional dysregulation becomes evident especially after 24 months of age and age-related post-transcriptional dysregulation leads to accumulation of core matrisome proteins and reduction of mitochondrial membrane proteins in multiple tissues. Based on our in-depth proteomic data and sample-matched transcriptome data of adult, middle-aged, old, and geriatric mice, we construct the Mouse aging proteomic atlas (<https://aging-proteomics.info/>), which provides a thorough and integrative view of age-related gene expression changes.

Aging is the biggest risk factor for most chronic diseases, and understanding its underlying mechanisms is pivotal for the development of preventive and curative treatments for age-related diseases. The first step in this process is a comprehensive and quantitative assessment of age-related molecular-level changes, and continuous efforts have been made to clarify transcriptome changes during aging<sup>1,2</sup>. Gene expression patterns can define cellular states, and transcriptomic analysis informs us of how gene expression levels, or more accurately mRNA levels, are affected by age. While transcriptomic information is fundamental to the understanding of the molecular processes of aging, it inevitably has some limitations. Changes in mRNA levels that are not

accompanied by protein abundance changes are likely to be inconsequential. In addition, protein levels may be regulated post-transcriptionally. Post-transcriptional regulation of protein levels not only consists of physiological intracellular mechanisms such as autophagy and the ubiquitin–proteasome system but also includes pathological or extracellular mechanisms such as ectopic accumulation of proteins delivered via blood circulation. Due to the pervasive influence of these pathways (hereafter collectively referred to as post-transcriptional regulation) on the functional gene expression profile, it is important to investigate gene expression changes at the protein level. Investigating protein levels is especially relevant for aging

<sup>1</sup>Department of Pathophysiology, Osaka Metropolitan University, Graduate School of Medicine, Osaka, Japan. <sup>2</sup>Proteomic Core Facility, EMBL Heidelberg, Heidelberg, Germany. <sup>3</sup>Laboratory of Proteomics for Drug Discovery, Center for Drug Design Research, National Institute of Biomedical Innovation, Health and Nutrition, Osaka, Japan. <sup>4</sup>Medical Research Support Center, Graduate School of Medicine, Kyoto University, Kyoto, Japan. <sup>5</sup>Department of Biology, University of Rochester, Rochester, NY, USA. <sup>6</sup>These authors contributed equally: Yoshiki Nonaka, Kazuaki Takemura, Yuya Yoshida. <sup>7</sup>These authors jointly supervised this work: Masaki Takasugi, Naoko Ohtani. ✉ e-mail: [mtjfc@gmail.com](mailto:mtjfc@gmail.com); [naoko.ohtani@omu.ac.jp](mailto:naoko.ohtani@omu.ac.jp)

studies since the loss of proteostasis is considered to be one of the major causes of aging<sup>3</sup>. However, although regulatory mechanisms of proteostasis have been implicated in the molecular processes of aging, age-related proteome changes themselves remain underexplored. Proteomic studies in the past uncovered age-related changes in the tissues such as brain, kidney, liver, and lung of mice<sup>4–7</sup>. These studies used up to 24-month-old C57BL/6 mice, although their maximum lifespan is considered to be around 36 months. In addition, recent cross-tissue analysis of age-related proteomic changes investigated 18-month-old C57BL/6 mice<sup>8</sup>, which are considered to be equivalent to the fifties in humans. Thus, it is unclear how the proteome changes after reaching old age and whether there are any common features in advanced aging proteome that apply to multiple tissues. To what extent age-related proteome changes are independent of local transcriptome, and the features of post-transcriptional dysregulation during aging, also remain to be elucidated.

Here, we conduct quantitative proteomic analyses of whole-tissue lysates (WTL) and low-solubility protein-enriched fractions (LSF) of major tissues collected from adult, middle-aged, old, and geriatric mice (6, 15, 24, and 30-month-old, respectively), and analyze these data in combination with transcriptomic data collected from the same mouse tissue samples. Here, we describe the major features of the mouse aging proteome and provide a useful resource for future aging studies (Mouse aging proteomic atlas, <https://aging-proteomics.info/>).

## Results

### Construction of a mouse aging proteomic atlas

While whole-tissue lysates (WTL) provide the least biased tissue proteome, the depth of proteomic profiling could be increased by additional analysis of fractionated samples or samples depleted of high-abundance proteins<sup>9–11</sup>. Thus, we added low-solubility protein-enriched fractions (LSF) that could be potentially underrepresented in WTL to the analysis (Supplementary Fig. 1a). To prepare WTL, we used 2 × Laemmli buffer containing a high concentration of SDS (4%) not only because it is an efficient detergent for protein extraction<sup>12</sup> but also because this is a standard lysate buffer used in protein expression analyzes and thus would provide a useful resource for researchers in related fields. Based on previous studies showing that 0.1% SDS wash can effectively enrich low-solubility proteins for proteomic analysis<sup>13,14</sup>, we prepared LSF by removing high-solubility proteins by pre-incubating tissues in low-concentration SDS solution (0.125 to 1%) followed by discarding the supernatant and lysing residual low-solubility protein-enriched materials in 2 × Laemmli buffer. After pre-incubation in low-concentration SDS, a substantial fraction of low-solubility proteins was recovered in heart, kidney, lung, gastrocnemius muscle, and abdominal skin but not in brain and liver (Supplementary Fig. 1b, c). To confirm the enrichment of low-solubility proteins, we performed proteomic analysis of these fractions using a Triple TOF 5600 LC–MS/MS system (Supplementary Fig. 1d). Although we did not aim to enrich a specific group of low-solubility proteins, we found that core matrisome and mitochondrial membrane proteins that are known to have low solubility in traditional buffer systems accounted for a higher proportion of proteins, peptides, and signals detected by mass spectrometry in LSF than in WTL (Supplementary Fig. 1e).

Based on these preliminary results, we investigated age-related proteome changes in LSF of heart, kidney, lung, muscle, and skin obtained by pre-incubating tissues in SDS solution at concentrations of 0.125% (heart, kidney, lung, and skin) and 0.25% (muscle). Note that there were no significant changes in the amount of proteins recovered in LSF over time in these tissues (Supplementary Fig. 1f). Proteins recovered in LSF of brain and liver obtained by pre-incubating tissues in 0.125% SDS solution accounted for only 1% of total proteins in WTL (Supplementary Fig. 1b), indicating that LSF of these tissues has very small contribution to the whole proteome, and thus although proteomic analyses of such minor aggregate fraction significantly

contributed to the better understanding of brain aging<sup>15–17</sup>, in light of our research aim we did not investigate age-related changes of brain and liver LSF in the current study. Regarding WTL, age-related proteome changes were investigated in thoracic aorta (LSF was not analyzed due to the small amount of tissue), brain, heart, kidney, liver, lung, muscle, and skin.

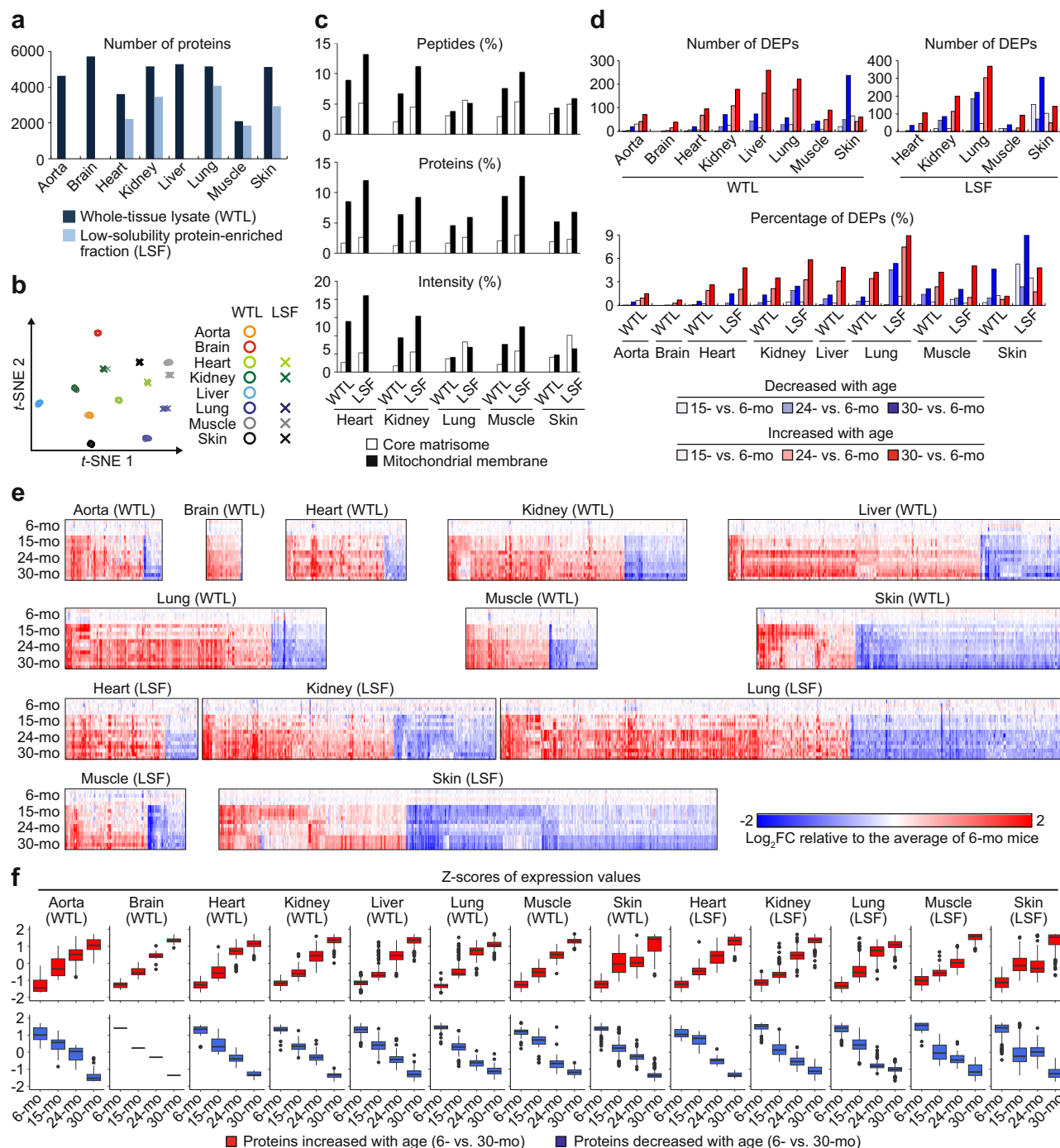
In-depth quantitative proteomic LC–MS/MS analysis was performed for these samples collected from 6-, 15-, 24-, and 30-month-old male C57BL/6J mice (4 biological replicates per age group) using 16-plex TMT labeling and an Orbitrap Fusion Lumos Tribrid mass spectrometer. For each tissue, WTL or LSF prepared from mice of different ages were multiplexed to enable accurate and quantitative comparisons between different age groups, i.e., all multiplexes were organized to enable comparisons between different age groups within either WTL or LSF of a single tissue. The layout of the TMT batches is shown in Supplementary Data 1 and the results of TMT-labeled proteomic analyzes are provided in Supplementary Data 2. We detected and quantified 9283 proteins in total and an average of 4608 and 2906 proteins in WTL and LSF, respectively (Fig. 1a). Overall, proteomic profiles of WTL and LSF of the same tissue were as different as those of different tissues (Fig. 1b). GO analysis showed that LSF-specific proteins were enriched with proteins in the organelle membrane and collagen-containing extracellular matrix (Supplementary Fig. 1g and Supplementary Data 3). In addition, proteins residing in the core matrisome and mitochondrial membrane, according to MatrisomeDB<sup>18</sup> and MitoCarta3.0<sup>19</sup> accounted for a higher proportion of proteins, peptides, and signals detected by mass spectrometry in LSF than in WTL (Fig. 1c and Supplementary Data 4), indicating the success of low-solubility protein enrichment and improvement of analytical comprehensiveness.

### Age-related changes in mouse tissue proteome

To take both statistical significance and biological relevance into consideration, the cut-off for differentially expressed proteins (DEPs) was set as FDR less than 0.05 with fold change greater than 1.5. DEPs identified with these criteria mostly had FDR less than 0.01 (Supplementary Data 5). The number of proteins significantly differentially expressed compared to 6-month-old mice showed a gradual increase with age (Fig. 1d). It is important to note, however, that the survival rate of C57BL/6J mice starts to drop at around 24 months of age and the effects of diseases on differential protein expressions could be significantly higher at 30 months than at 24 months of age. That being said, most of the age-dependent proteins exhibited age-progressive changes in their expression, although in the skin a considerable portion of age-dependent proteins exhibited transient expression changes in 15-month-old mice (Fig. 1e). The majority of proteins significantly differentially expressed in 30-month-old mice compared to 6-month-old mice showed gradual age-related changes (Fig. 1f and Supplementary Data 6), suggesting that these changes are associated with natural aging process. Many changes associated with diseases may tend not be detected with statistical significance due to their lower prevalence compared to aging. Expression levels of cellular senescence marker *p16* and DNA damage maker  $\gamma$ -H2AX were also higher at 30 months than at 24 months of age (Supplementary Figs. 2a, b). In order to study developed and comprehensive age-related changes, we decided to further characterize the differences in proteome between 6- and 30-month-old mice.

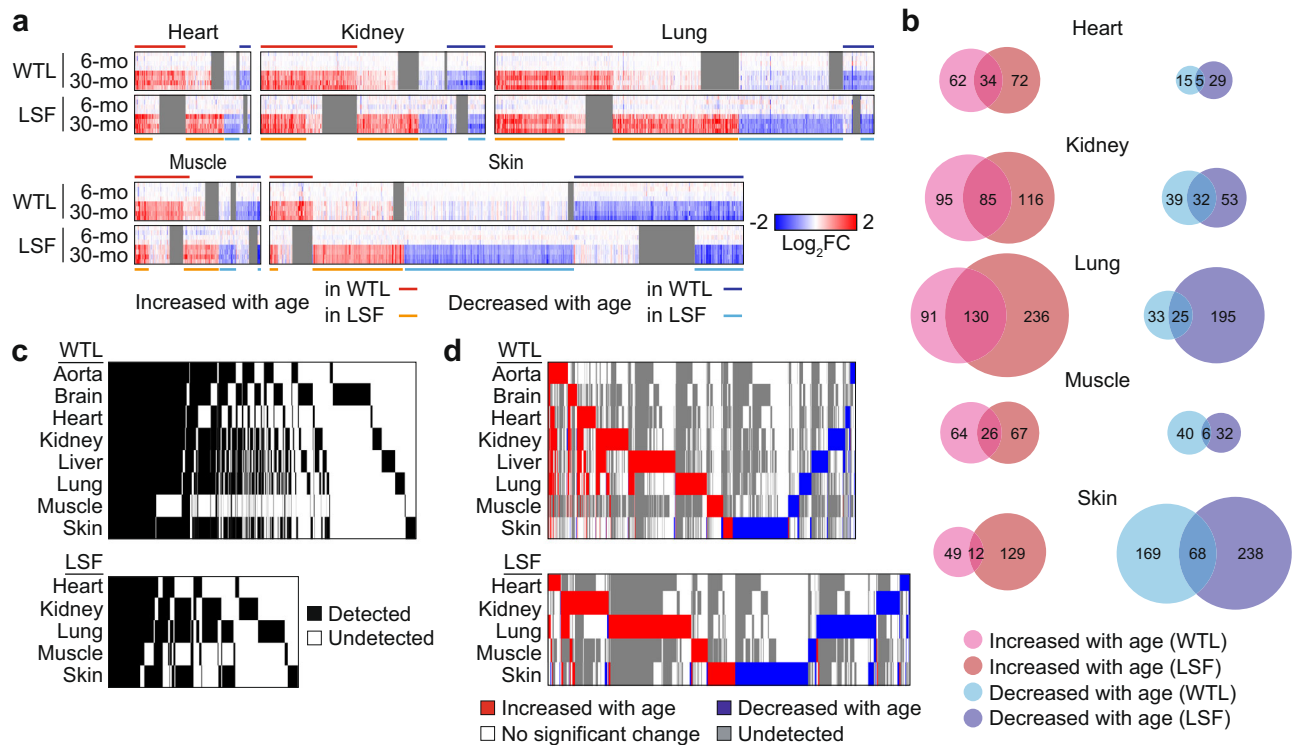
At 30 months of age, the percentage of DEPs in WTL was 5–6% in kidney, liver, lung, muscle, and skin, 2–3% in aorta and heart, and 0.7% in the brain. The small number of DEPs in the brain may be due to the complexity of the brain tissue or may suggest that the brain has less resistance to gene expression dysregulation and has to be better preserved during aging.

The percentages of DEPs were higher in LSF, and were 6–8% in heart, kidney, and muscle, and 14–15% in lung and skin (Fig. 1d).



**Fig. 1 | Identification of proteins differentially expressed with age in whole-tissue lysates (WTL) and low-solubility protein-enriched fractions (LSF) of major mouse organs.** **a** The number of proteins detected and quantified by proteomic mass spectrometry analysis using 16-plex TMT labeling and an Orbitrap Fusion Lumos Tribrid mass spectrometer. **b** tSNE plot of the proteomic profiles of WTL and LSF of major organs constructed using the Seurat pipeline. The intensities of proteins normalized within the multiplexed dataset were used for the analysis. Different colors represent different tissues and ○ and × represent WTL and LSF, respectively. **c** The top bar graph shows the percentage of peptides of core matrisome and mitochondrial membrane proteins within the total peptides measured in WTL and LSF. The middle bar graph shows the percentage of core matrisome and mitochondrial membrane proteins within the total identified proteins. The bottom bar graph shows the percentage of signal intensity derived from core matrisome and mitochondrial membrane proteins in the total signal intensity. **d** The top two graphs show the number of proteins differentially expressed with

age in WTL and LSF of major mouse organs. Differentially expressed proteins (DEPs) were defined as those with a fold change > 1.5 and FDR < 0.05. The bottom graph shows the percentage of DEPs within the total identified proteins. **e** The heatmaps show the relative protein expression levels of proteins differentially expressed between 6- and 15-, 24-, or 30-month-old mice. The log<sub>2</sub> fold changes are relative to the average of 6-month-old mice. The color scheme shows the log<sub>2</sub> fold change. The log<sub>2</sub> fold changes above 2 were truncated to 2. **f** The boxplots show the Z-scores of expression values of proteins differentially expressed between 6- and 30-month-old mice. Expression levels of each protein were averaged in each age group and then z-scored across all age groups. The top and bottom graphs show the expression Z-score of proteins that increase and decrease with age, respectively. Central lines represent median values and box limits represent upper and lower quartiles. Whiskers represent 1.5 × the interquartile range above and below the upper and lower quartiles, respectively.



**Fig. 2 | Tissue- and fraction-dependent changes in the proteome during aging.**

**a** The heatmaps show the relative protein expression levels of proteins differentially expressed between 6- and 30-month-old mice in either or both WTL and LSF. Gray indicates that the corresponding protein was not detected in that fraction. The log<sub>2</sub> fold changes are relative to the average of 6-month-old mice. Note that expression fold change values themselves can be only compared among a given

column (given protein) of a given heatmap. **b** The Venn diagrams show the overlap between proteins differentially expressed between 6- and 30-month-old mice in WTL and LSF. All DEPs identified in WTL or LSF were considered. **c** The heatmaps show the overlaps among proteins detected in different tissues. **d** The heatmaps show the overlaps among proteins differentially expressed with age (6- vs. 30-months) in different tissues.

Analysis of LSF doubled the number of DEPs (Fig. 2a, b). The majority of DEPs found in LSF were also detected in WTL, but were not differentially expressed with age in WTL, suggesting that these proteins were affected by age only in specific cellular compartments or when associated with other conditions that decrease their solubility (Fig. 2a).

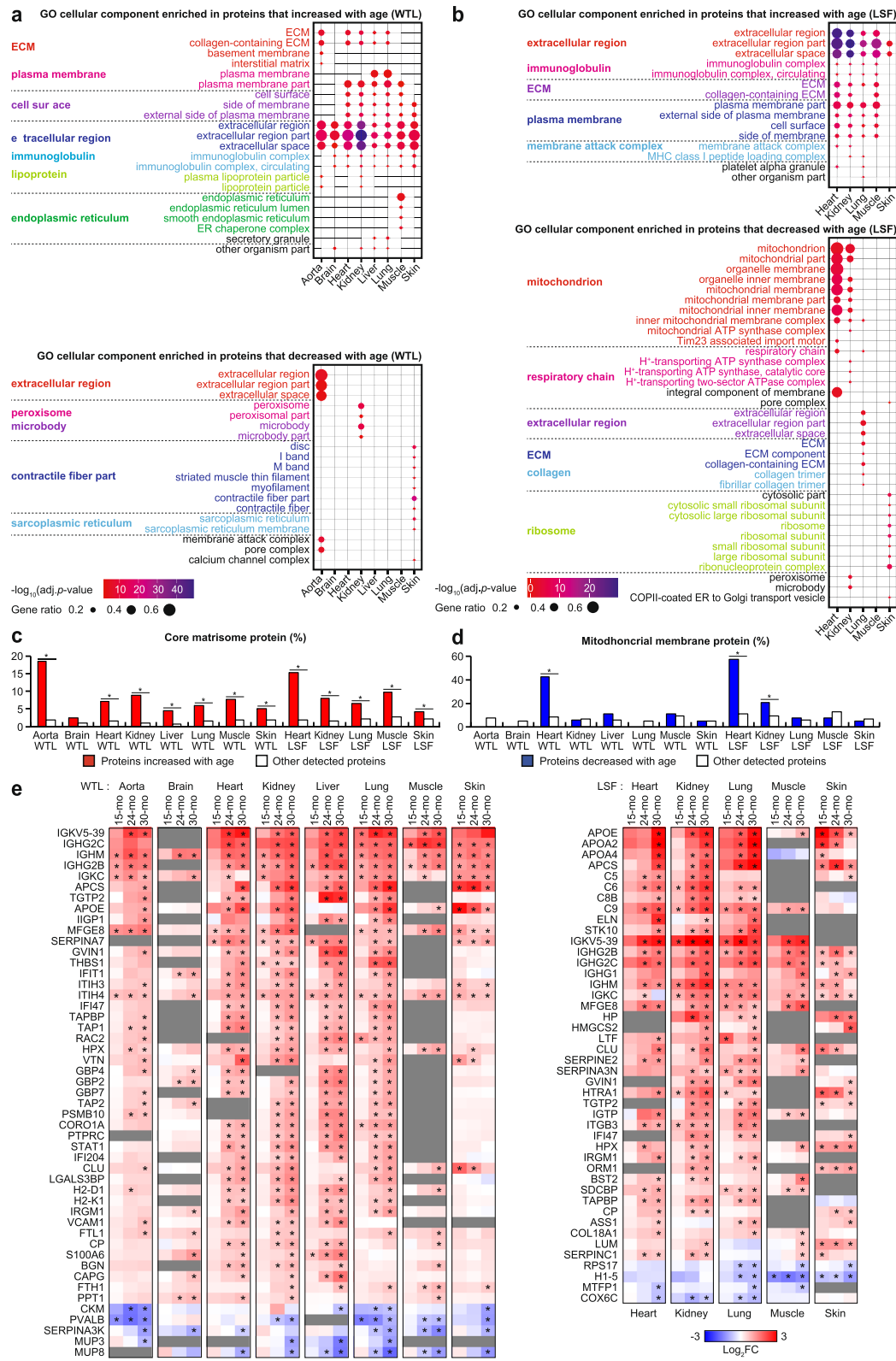
There was significant overlap among proteins detected in each tissue (Fig. 2c and Supplementary Data 7), but the vast majority of DEPs were tissue-dependent (Fig. 2d and Supplementary Data 8). However, the top GO terms enriched in the DEPs of each tissue extensively overlapped with each other (Figs. 3a, b, Supplementary Fig. 3a, b, and Supplementary Data 9). Proteins associated with the immune response preferentially increased with age in all tissues in WTL and LSF (Supplementary Figs. 3a, b). Proteins increased with age specifically in LSF also showed an enrichment of immune response proteins (Supplementary Fig. 4). Extracellular matrix (ECM) proteins increased with age in aorta, heart, kidney, liver, lung, and muscle in WTL and LSF (Figs. 3a, b). Proteins commonly increased with age in both WTL and LSF and proteins increased with age specifically in WTL or LSF were all enriched in ECM proteins (Supplementary Fig. 5). Core matrisome proteins retrieved from Matrisome DB<sup>18</sup> were also enriched in proteins that increased with age in all the tissues regardless of the sample type and the enrichment was statistically significant in 12 out of 13 datasets (Fig. 3c). Proteins in the mitochondrion or inner mitochondrial membrane complex were disproportionately decreased with age in heart, kidney, and lung in LSF but not in WTL (Fig. 3b). Mitochondrial membrane proteins retrieved from MitoCarta3.0<sup>19</sup> were also significantly enriched in proteins that decreased with age in heart and kidney (Fig. 3d). Importantly, although core matrisome and mitochondrial membrane proteins were enriched in DEPs in multiple tissues, most of these DEPs were tissue-dependent, indicating that these compartments significantly change during aging in multiple tissues

but in a highly tissue-dependent manner. DEPs associated with the immune response were more common across tissues, but still a substantial proportion of them were tissue-specific in kidney, liver, lung, and skin (Supplementary Fig. 6a and Supplementary Data 10).

In addition to commonly enriched GO terms, several interesting GO terms were enriched in DEPs in a specific tissue. For example, in the aorta, proteins that decreased with age were enriched with complement activation-related proteins. In contrast, in all the other tissues, these proteins were enriched in proteins that were increased with age (Supplementary Fig. 3a). In the muscle, proteins increased with age were enriched with proteins in the endoplasmic reticulum (ER) and ER chaperone complex (Fig. 3a).

As exemplified by the immune response-related genes, a fraction of proteins were commonly affected by age across different tissues. For each WTL and LSF, we defined proteins that were significantly and concordantly increased or decreased in geriatric mice in half or more of the investigated tissues as common DEPs (Fig. 3e). Common DEPs that increased with age were most enriched with proteins associated with KEGG pathways “Antigen processing and presentation (adjusted  $p$ -value =  $1.1 \times 10^{-4}$ )” and “Complement and coagulation cascades (adjusted  $p$ -value =  $3.7 \times 10^{-5}$ )” in WTL and LSF, respectively. The plasma proteome of aged mice reported by two recent studies<sup>20,21</sup> did not show similar age-related expression changes in common DEPs, with a few exceptions such as VCAM1 and Lactoferrin (LTF) (Supplementary Fig. 6b). Rather, the majority of common DEPs exhibited concordant changes in mRNA levels (Supplementary Fig. 6c), which is consistent with a previous transcriptomic study reporting an increased infiltration of immune cells in aged tissues<sup>4</sup>. We harvested the mouse tissue samples without prior perfusion or blood collection and thus age-related changes in blood flow or immune cells in the tissues may be sources of the immune-related common DEPs. Commonly





increased proteins that were predominantly expressed in non-immune cells included SERPINE2 and COL18A1 (Fig. 3e). SERPINE2 and C-terminal fragments of COL18A1 are known to have anti-angiogenic effects, and age-related increases in SERPINE2 levels have been reported to be involved in impaired angiogenesis in the aged mouse heart<sup>22</sup>. Common DEPs that were decreased with age were relatively small in number and included creatine kinase (CKM), parvalbumin

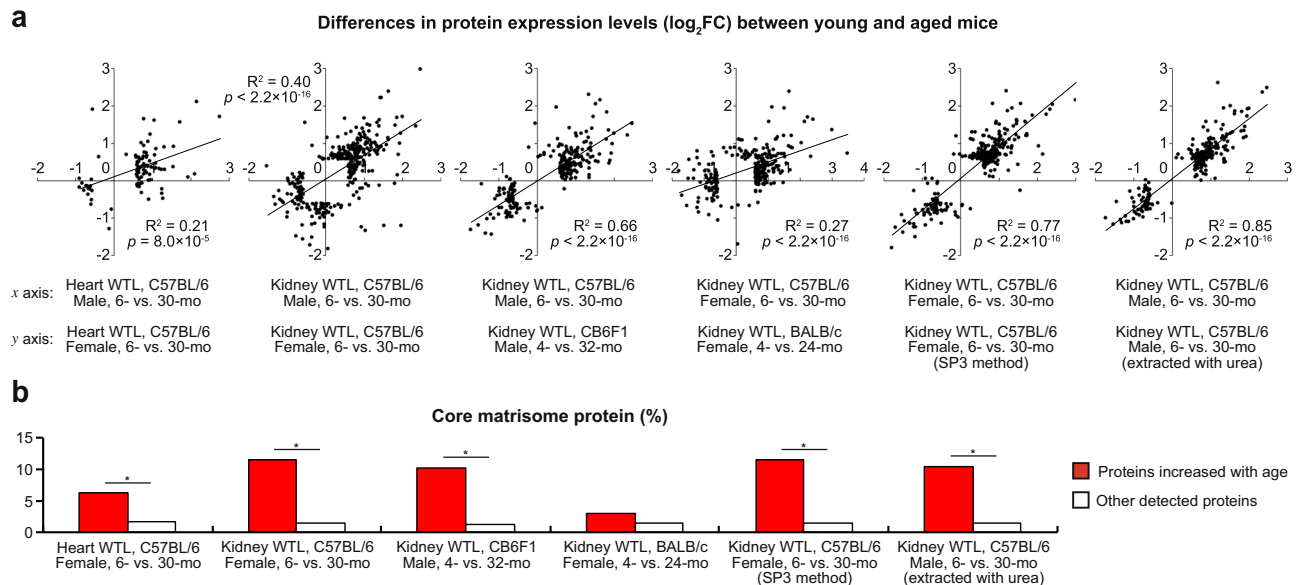
(PVALB), major urinary proteins (MUP3 and MUP8), ribosomal protein S17 (RPS17), linker histone H1.5 (H1-5), mitochondrial fission process 1 (MTFP1), and COX6C (Fig. 3e).

### Effects of sex and genetic background

Age-related mRNA expression changes have been reported to be significantly different between male and female in mouse<sup>1</sup> and human<sup>23</sup>.

**Fig. 3 | Functional characteristics of DEPs.** **a, b** The dot plots show GO cellular component terms that were enriched in DEPs compared to the total detected proteins (adjusted  $p$ -value  $< 0.05$ ). The top 10 significantly enriched terms with the lowest adjusted  $p$ -values were determined for each tissue and then combined with those of other tissues. These terms were then clustered based on simplify Enrichment R package. **a** GO term enrichment in proteins differentially expressed in WTL of 6- and 30-month-old mice. **b** GO term enrichment in proteins differentially expressed in LSF of 6- and 30-month-old mice. **c** The proportion of core matrisome proteins in proteins that increased with age and in other detected proteins. **d** The proportion of mitochondrial membrane proteins in proteins that decreased with age and in other detected proteins. **e** The heatmaps show the age-related changes

in the protein levels of common DEPs. Common DEPs are defined here as proteins whose levels significantly and concordantly increased or decreased in 30-month-old mice compared to 6-month-old mice in half or more of the investigated tissues. Log<sub>2</sub> fold changes are relative to the average of 6-month-old mice. Gray indicates that the corresponding protein was not detected in that tissue. The left and right heatmaps show common DEPs found in WTL and LSF, respectively. Note that expression fold change values themselves can be only compared among a given row (given protein) of a given heatmap. \* Adjusted  $p$ -value  $< 0.05$  [Fisher's exact test corrected by BKJ method for (c, d) and fold change cut-off of 1.5 and limma FDR  $< 0.05$  for (e)].



**Fig. 4 | Evaluation of the generality of the findings.** **a** Differences in protein expression levels between young and old mice measured under one condition are plotted against those measured under another condition. Differential expression levels are shown for all DEPs identified in either or both of the conditions shown

under each plot. Results of two-tailed Spearman correlation tests are shown on the plots. **b** The proportion of core matrisome proteins in proteins that increased with age and in other detected proteins. \* Adjusted  $p$ -value  $< 0.05$  (two-tailed Fisher's exact test corrected by BKJ method).

Genetic background also has been reported to have considerable effects on age-related transcriptomic changes<sup>24</sup>. In order to evaluate to what extent the findings from the proteomic study of young and old C57BL/6J male mice can be generalized to female mice and mice of other genetic backgrounds, we performed proteomic analysis of WTL of heart and kidney of 6- and 30-month old female C57BL/6J mice ( $n = 4$  with 8-plex TMT labeling) and WTL of the kidney of 4- and 32-month-old CB6F1 male mice and 4- and 24-month-old BALB/c female mice ( $n = 5$  using 10-plex TMT labeling) (Fig. 4a). Age-related expression changes in the protein levels of DEPs identified in either or both sexes showed a good correlation between male and female C57BL/6J mice. Overall age-related proteomic changes were also similar among C57BL/6J, CB6F1, and BALB/c mice. In addition, analysis of WTL extracted with 8 M urea buffer instead of 4% SDS buffer, and of WTL-derived peptides prepared using single-pot solid-phase-enhanced sample preparation (SP3) method<sup>25</sup> instead of S-Trap column (that was used in all the other preparations) ( $n = 4$  with 8-plex TMT labeling) showed similar age-related proteome changes in the kidney of C57BL/6J mice. That is, proteins significantly differentially expressed with age in one of these of alternative measurements or in its control measurement mostly showed consistent age-related protein expression changes in both measurements (Fig. 4a). Enrichment of core matrisome proteins in proteins increased with age, a prominent tissue-independent feature of the aging proteome in male C57BL/6J mice (Fig. 3c), was observed in all the above conditions, although without statistical significance in female BALB/c mouse kidney (Fig. 4b). It should be noted, however, that although there was an overall

correlation in age-related changes between different conditions, a considerable number of proteins were significantly differentially expressed with age only in a specific condition. For example, DEPs decreased with age in male but not in female C57BL/6 mice kidney were enriched with microbody- and peroxisome-associated proteins. On the other hand, DEPs decreased with age in female but not in male C57BL/6 mice kidney were enriched with basolateral plasma membrane- and spectrin-associated cytoskeleton-associated proteins (Supplementary Data 11).

### Features of age-related post-transcriptional dysregulation

We next sought to investigate the concordance between age-related changes in transcriptome and proteome. To avoid the effects of inter-individual differences and the issues arising from sampling timing and methods, we performed transcriptomic analysis of brain, heart, kidney, liver, lung, muscle, and skin of 6, 15, 24, and 30-month-old male mice using the same frozen tissue powder used for the proteomic analyzes. Due to its small size, the whole thoracic aorta was directly lysed in  $2 \times$  Laemmli buffer and thus we conducted transcriptomic analysis of aorta collected from other mice. Differentially expressed mRNAs (DERs) were determined using the same criteria as for DEPs (fold change cut-off of 1.5 and FDR  $< 0.05$ ). mRNAs identified to be differentially expressed between 6 and 24-month-old C57BL/6 male mice in our dataset were considerably different from those identified in the *Tabula Muris Senis* dataset<sup>1</sup> (Supplementary Fig. 7a), confirming the importance of minimizing artificial discrepancies in multi-omics analysis. For reference, genes whose expression was consistently

different between 6 and 24-month-old mice in our proteome and transcriptome data and in the *Tabula Muris Senis* data, which could serve as reliable markers of mouse aging, are shown in Supplementary Fig. 7b.

A combination of our transcriptomic and proteomic data of 6- and 30-month-old mouse tissues revealed that only up to approximately 60% of DERs and DEPs accompanied significant concordant changes in the levels of their corresponding protein and mRNA, respectively (Fig. 5a, b). Proteins that increased during aging in kidney and liver were exceptions and 69–80% of them were significantly upregulated at the mRNA level (Fig. 5b). The concordance between age-related changes in the levels of mRNA and protein was lower in downregulated genes than in upregulated genes. Given that a considerable portion of DEPs could be affected by post-transcriptional mechanisms, we next sought to characterize the features of DEPs that were not accompanied by significant changes in mRNA levels. As a result, we found that genes with significantly increased expression only at the protein level had higher ratios of core matrisome genes than those that had increased expression at both the mRNA and protein levels in 12 out of 13 datasets (Fig. 5c). On the other hand, genes with significantly decreased expression only at the protein level had higher ratios of mitochondrial membrane genes than those that had decreased expression at both the mRNA and protein levels in 10 out of 13 datasets (Fig. 5c). Both of these trends were statistically significant in our datasets (Supplementary Fig. 7c). Non-membrane mitochondrial genes were not enriched in genes that decreased with age only at the protein level (Fig. 5c). Mitochondrial membrane proteins that decreased with age only at the protein level were enriched with OXPHOS proteins in most datasets, with statistical significance ( $FDR < 0.05$ ) in LSF of kidney and lung (Fig. 5d).

We also found in 11 out of 13 datasets that proteins differentially expressed in 30-month-old mice without accompanying significant concordant changes in mRNA levels were less likely to be significantly differentially expressed by 24 months of age (Fig. 5e). This finding suggests that the overall contribution of post-transcriptional mechanisms to age-related changes in protein abundance generally becomes higher after reaching old age.

To create a catalog of genes that are dysregulated in very old age primarily via post-transcriptional mechanisms, we listed genes whose protein product significantly increased with age (i.e. fold change cut-off of 1.5 and  $FDR < 0.05$ ) without accompanying greater than 1.05-fold increase in their corresponding mRNA levels, and genes whose protein product significantly decreased with age without accompanying greater than 1.05-fold decrease in their corresponding mRNA levels (Fig. 6 and Supplementary Fig. 8). A total of 198 and 502 DEPs found in WTL and LSF met these criteria, respectively. Altogether there were a total of 635 such proteins, including 41 core matrisome proteins and 36 mitochondrial membrane proteins. Ferritin heavy chain (FTH1) was post-transcriptionally increased in WTL of kidney, liver, and skin, and ferritin light chain (FTL1) was post-transcriptionally increased in WTL of aorta and kidney and in LSF of kidney and skin, suggesting that post-transcriptional mechanisms affect iron homeostasis in very old age. We also found that HTRA1, an extracellular protease whose increased expression is known to induce extracellular protein deposition<sup>26</sup>, was post-transcriptionally increased in WTL of aorta and lung and in LSF of heart, lung, and skin. It should be noted, however, that the mechanisms underlying the discrepancy between age-related changes in mRNA and protein levels remain unclear. Expressions of core autophagy genes<sup>27</sup> showed a trend of decrease with age at both the mRNA and protein level in the liver (Supplementary Fig. 9), a tissue in which age-related decline in autophagy activity has been well demonstrated<sup>28,29</sup>. Whether proteostasis regulators including autophagy explain the discrepancy between age-related changes in mRNA and protein levels cannot be concluded at this point and warrants further investigation.

## Cellular source of DEPs

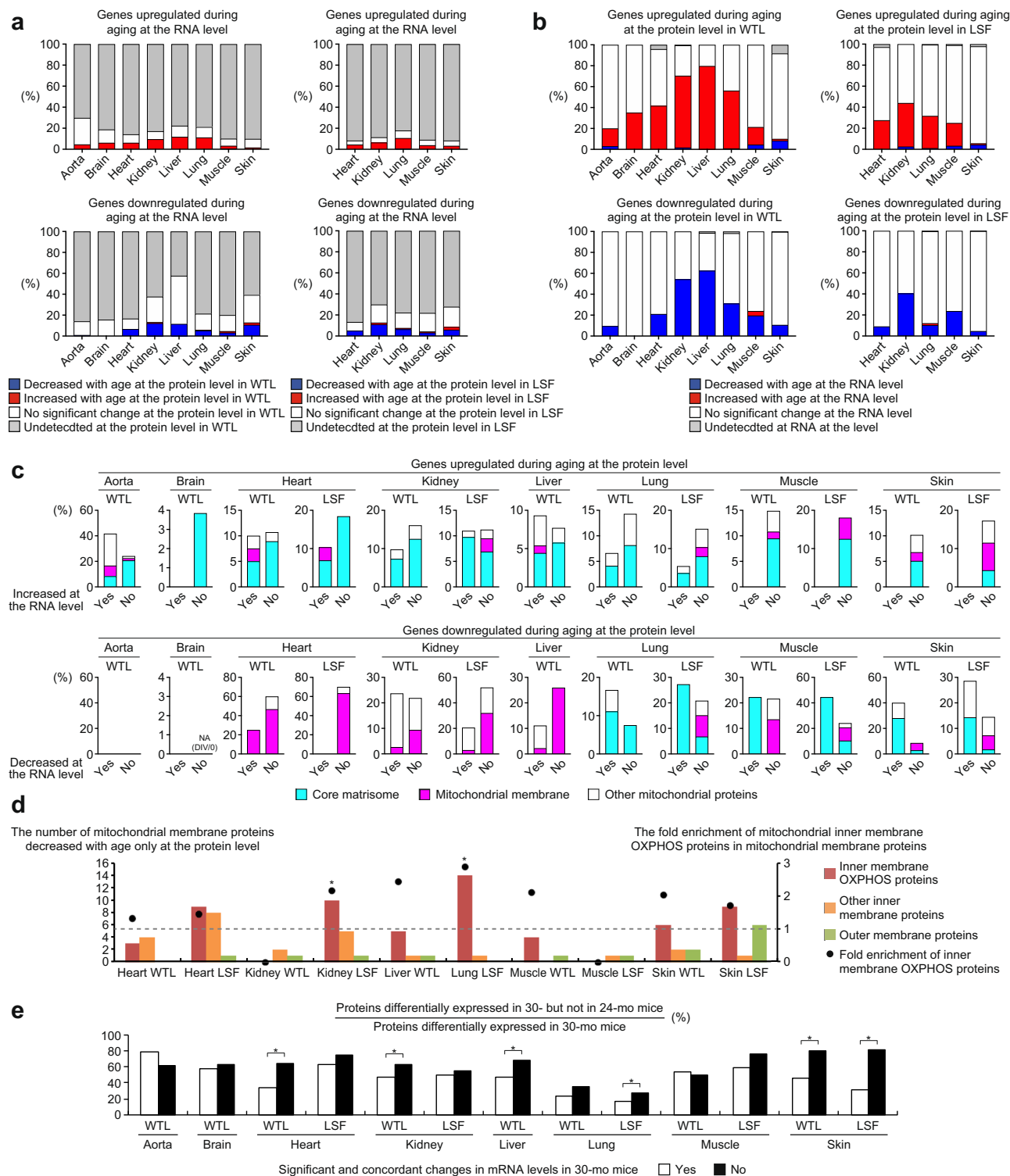
To investigate the cellular source of DEPs, we next examined the mRNA levels of DEPs at the single-cell level using the public single-nucleus RNA-Seq data of normal adult male C57BL/6 mouse heart<sup>30</sup>, kidney<sup>31</sup>, liver<sup>32</sup>, lung<sup>33</sup>, and muscle<sup>34</sup>. Brain was not included in the analysis due to the low number of DEPs. Aorta and skin were excluded due to the lack of public single-nucleus RNA-Seq data. A cell type expressing a DEP-encoding mRNA at high levels is likely to contribute to the expression of that DEP in the tissue. If expressing a DEP-encoding mRNA at the same levels, more abundant cell type is likely to have higher contribution to the expression of that DEP in the tissue.

Analysis of public single-nucleus RNA-Seq data suggested that proteins increased with age and proteins decreased with age and were produced by different types of cells (Fig. 7 and Supplementary Data 12). Genes with increased expression levels in 30-month-old mice at both the protein and mRNA levels tend to be highly expressed in immune cells (the leftmost column of Fig. 7), and core matrisome proteins increased with age tend to be highly expressed in stromal cells (the third column from the left in Fig. 7). On the other hand, proteins with decreased expression levels in 30-month-old mice tend to be highly expressed in parenchymal cells (the right three columns of Fig. 7). For example in the liver, proteins increased with age at both the protein and mRNA levels were abundantly expressed in immune cells, and core matrisome proteins increased with age were abundantly expressed in stellate cells. In contrast, almost all of the proteins decreased with age in the liver were predominantly expressed in hepatocytes.

Regarding other tissues, proteins increased with age at both the protein and mRNA levels were most abundantly expressed in immune cells in heart and lung (the leftmost column of Fig. 7). This finding is consistent with our observations that common DEPs that increased with age were enriched with immune-related proteins (Fig. 3e) and tended to be differentially expressed at the mRNA level (Supplementary Fig. 6c). Core matrisome proteins that increased with age were highly expressed at the mRNA level in endocardial cells, fibroblasts, pericytes, and smooth muscle cells in the heart, fibroblasts, juxtaglomerular cells, mesangial cells, and podocytes in the kidney, and fibro-adipogenic progenitors, smooth muscle cells, and tenocytes in the muscle (the third column from the left in Fig. 7). Genes with decreased expression levels in 30-month-old mice at both the protein and mRNA levels were highly expressed in proximal tubular cells in the kidney (the fourth column from the left in Fig. 7). Genes with decreased expression levels in 30-month-old mice at the protein but not mRNA level were highly expressed in adipocytes, cardiomyocytes, and schwann cells in the heart, proximal tubular cells in the kidney, and skeletal muscle cells in the muscle (the fifth column from the left in Fig. 7). Mitochondrial membrane proteins that decreased with age were also highly expressed in these cells (the rightmost columns of Fig. 7). It is important to note that age-related changes of protein levels that were accompanied by changes in mRNA levels could be at least partially due to changes in cell population within a tissue, especially increase in immune cells and depletion of parenchymal cells.

## Validation of DEPs

Finally, we sought to validate the DEPs identified by our mass spectrometry proteomics. We first focused on APOE, not only because its protein levels were increased during aging in many tissues but also because human genetic polymorphisms that have the highest impact on lifespan and healthspan are those of APOE gene<sup>35</sup>. APOE is mainly produced in liver and brain and it is widely believed that its polymorphisms affect lifespan through its effects on Alzheimer pathogenesis, although recent study showed that APOE polymorphisms are associated with human longevity irrespective of Alzheimer's disease status<sup>36</sup>. Interestingly, our proteomic data showed that APOE was increased during aging in aorta, heart, kidney, lung, muscle, and skin



but not in brain and liver (Fig. 3e). We confirmed this finding by Western blotting (Fig. 8a). Real-time PCR showed a significant age-related increase in *ApoE* mRNA levels only in brain and muscle (Fig. 8b). Immunofluorescence staining showed that APOE was increased in the extracellular region in the aged heart (Figs. 8c, d), suggesting the extracellular deposition of circulating APOE. It is worth mentioning that a recent proteomic study of young and old human heart left ventricles also showed age-related increases in APOE and many other ECM proteins<sup>37</sup>. Considering that the liver is the major source of circulating APOE, altered tissue metabolism of APOE might be a cause of the discrepancy between age-related changes in *ApoE* mRNA and APOE

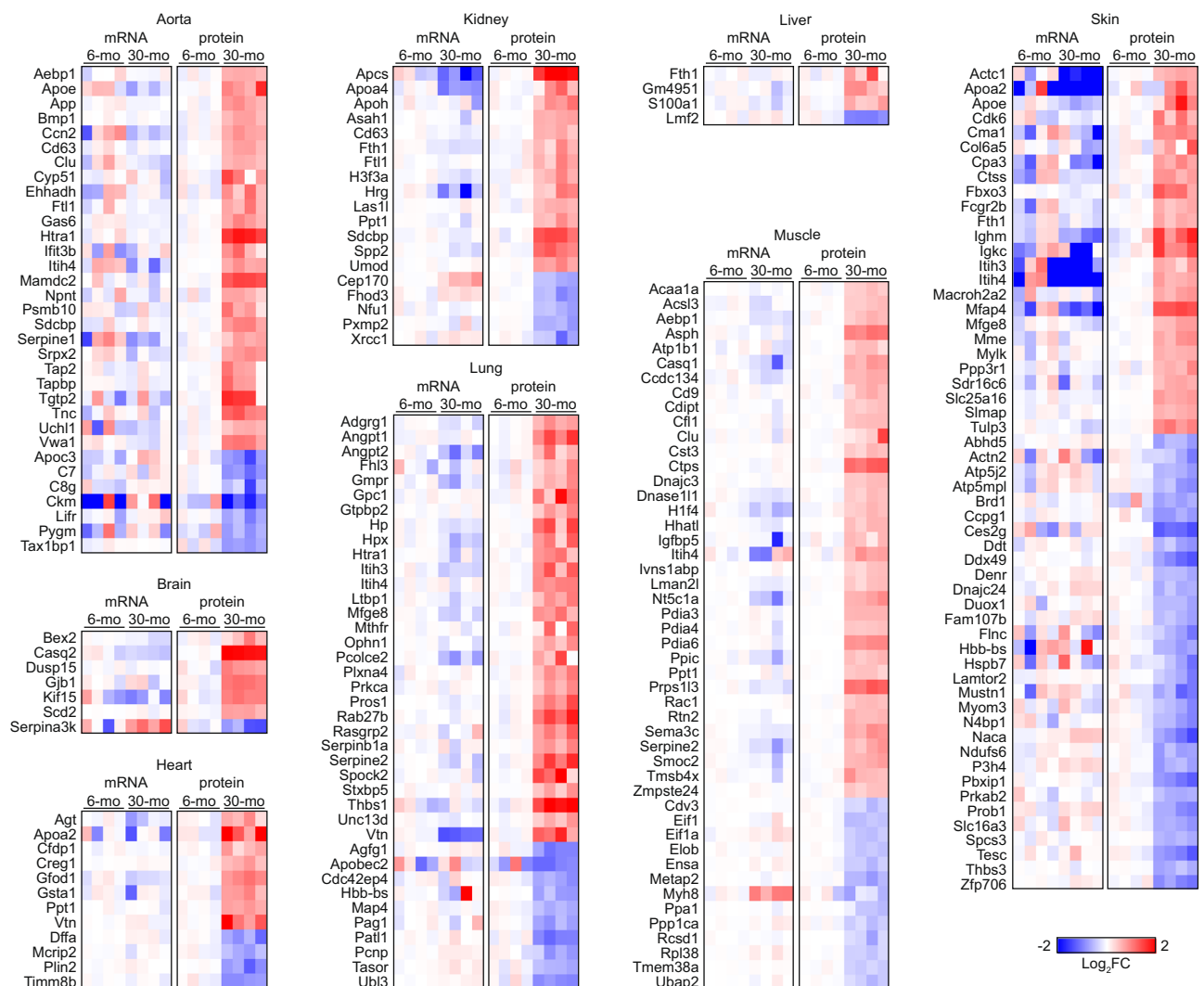
protein levels in aorta, heart, kidney, lung, and skin. Besides APOE has aggregation prone sequences<sup>38</sup> and thus age-related increase in APOE protein levels might be due to the accumulation of aggregated APOE. Another interesting common DEP was COL18A1 (Fig. 3e), whose C-terminal cleavage product known as endostatin has potent anti-angiogenic activity<sup>39</sup>. Western blotting showed that the ~26 kDa endostatin fragment was increased in aged mice in aorta, heart, kidney, lung, and muscle (Supplementary Fig. 10).

We also validated age-dependent expression of HSPA5 (Figs. 8e, f), a tissue-specific DEPs that was increased with age only in the muscle. A recent proteomic study of 8- and 18-month-old mice



**Fig. 5 | Features of age-related post-transcriptional dysregulation.** **a** The bar graphs show the proportion of differentially expressed mRNAs (DERs) that were undetected (gray) or detected with (red = increased with age; blue = decreased with age) or without (white) significant age-related changes by proteomic mass spectrometry analysis. Comparisons were made between 6- and 30-month-old mice. The criteria of DERs were identical to those of DEPs, which were fold change cut-off of 1.5 and FDR < 0.05. **b** The bar graphs show the proportion of DEPs that were undetected (gray) or detected with (red = increased with age; blue = decreased with age) or without (white) significant age-related changes by bulk RNA-Seq analysis. **c** The bar graphs show the proportion of core matrisome, mitochondrial membrane, and other mitochondrial proteins in the total number of proteins differentially expressed between 6- and 30-month-old mice. The proportion was calculated separately for DEPs that were accompanied by significant and concordant changes in mRNA levels and DEPs that were not accompanied by significant and concordant

changes in mRNA levels. **d** The bar graph shows the number of mitochondrial membrane proteins that decreased with age (6- vs. 30-month-old mice) without accompanying significant and concordant changes in mRNA levels. The numbers are presented separately for OXPHOS proteins, other inner membrane proteins, and outer membrane proteins. The dots show the fold enrichment of mitochondrial inner membrane OXPHOS proteins in mitochondrial membrane proteins that decreased with age without accompanying significant and concordant changes in mRNA levels. **e** The bar graph shows the proportion of proteins that were not differentially expressed in 24-month-old mice in proteins that were differentially expressed in 30-month-old mice. The proportion was calculated separately for DEPs that were accompanied by significant and concordant changes in mRNA levels and DEPs that were not accompanied by significant and concordant changes in mRNA levels. \* Adjusted *p*-value < 0.05 (two-tailed Fisher's exact test corrected by BKY method).

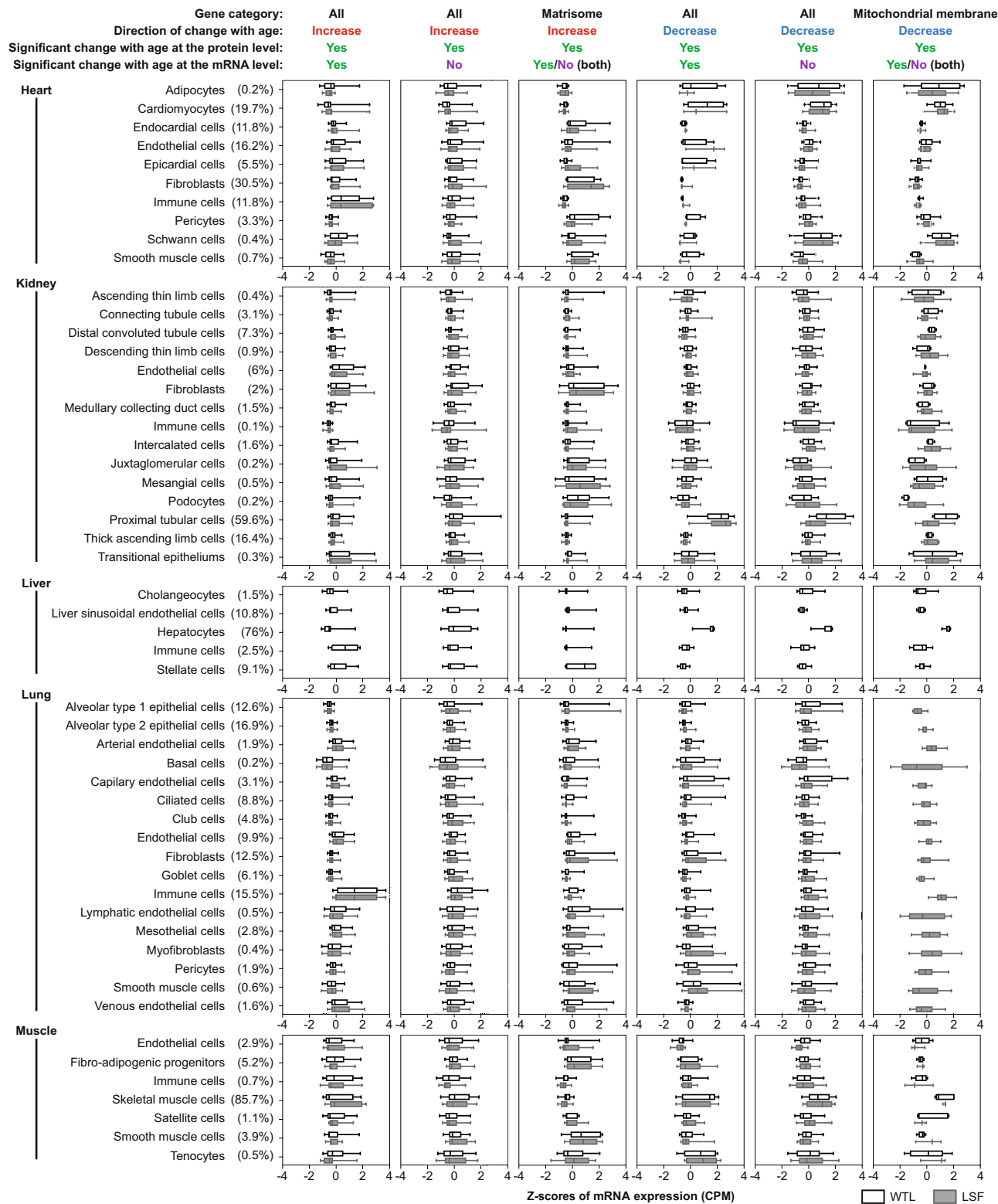


**Fig. 6 | Proteins differentially expressed with age in WTL that were not accompanied by similar changes in mRNA levels.** The heatmaps show relative expression levels of genes whose protein products were significantly differentially expressed in WTL between 6- and 30-month-old mice (i.e. fold change cut-off of 1.5 and FDR < 0.05). Log<sub>2</sub> fold changes are relative to the average of 6-month-old mice. Only genes whose protein product significantly increased with age without

accompanying greater than 1.05-fold increase in their corresponding mRNA levels, and genes whose protein product significantly decreased with age without accompanying greater than 1.05-fold decrease in their corresponding mRNA levels, are shown in the figure. The left and right sides of each heatmap show relative expression levels in mRNA and protein levels, respectively.

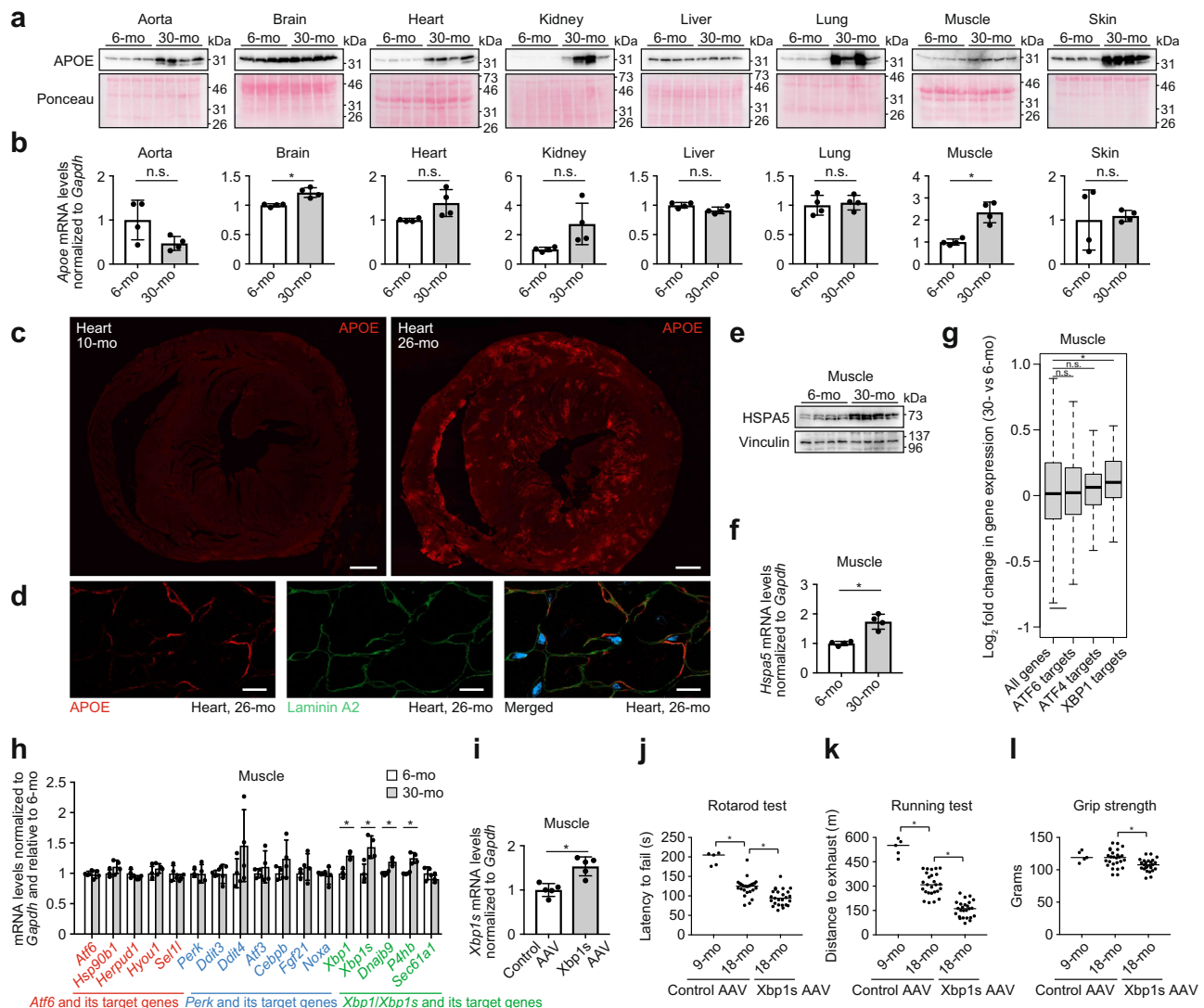
tissues also showed that HSPA5 specifically increases in aged male mouse muscle<sup>8</sup>. Upregulation of HSPA5 caught our attention as it is a marker of unfolded protein response (UPR) that is regulated by ER stress-activated transcription factors ATF6 and XBP1<sup>40</sup>. This, together

with the fact that the enrichment of ER/ER chaperone-proteins in the proteins that increased with age was observed only in the muscle (Fig. 3a), suggests the occurrence of UPR, which has been reported to modulate aging processes<sup>41</sup>. We examined the age-related expression



**Fig. 7 | Cell-type dependent mRNA expression levels of DEPs.** The boxplots show cell-type dependent mRNA expression levels of DEPs indicated at the top of the figure. mRNA expression data (CPM) were retrieved from public snRNA-Seq datasets. For each gene, expression was first averaged by cell type and then z-scored across all cell types within a tissue. Gene sets indicated at the top of the figure are 1) genes with increased expression in 30-month-old mice compared to 6-month-old mice according to both mass spectrometry and bulk RNA-Seq data; 2) genes with increased expression in 30-month-old mice compared to 6-month-old mice according to mass spectrometry but not bulk RNA-Seq data; 3) core matrisome genes with increased protein expression in 30-month-old mice compared to 6-

month-old mice; 4) genes with decreased expression in 30-month-old mice compared to 6-month-old mice according to both mass spectrometry and bulk RNA-Seq data; 5) genes with decreased expression in 30-month-old mice compared to 6-month-old mice according to mass spectrometry but not bulk RNA-Seq data; and 6) mitochondrial membrane genes with decreased protein expression in 30-month-old mice compared to 6-month-old mice. Central lines represent median values, and box limits represent upper and lower quartiles. Whiskers indicate the 10–90 percentiles. The numbers in the parenthesis indicate the ratio of the nucleus of each cell type detected by snRNA-Seq.



**Fig. 8 | Validation of DEPs identified by proteomic analysis.** **a** Western blots of APOE in 6- and 30-month-old mice WTL ( $n = 4$ ). **b** *ApoE* mRNA levels in 6- and 30-month-old mice measured by real-time PCR ( $n = 4$ ). **c** Representative immunofluorescence images of APOE in the hearts of 10- and 26-month-old mice. Five 10-month-old mice and five 26-month-old mice were analyzed with similar results. Scale bar = 500  $\mu$ m. **d** Representative confocal microscopy images of APOE and laminin a2 in the hearts of 26-month-old mice. Eleven mice were analyzed with similar results. Scale bar = 10  $\mu$ m. **e** Western blots of HSPA5 in 6- and 30-month-old mice. **f** *Hspa5* mRNA levels in 6- and 30-month-old mice measured by real-time PCR ( $n = 4$ ). **g** The boxplot shows  $\log_2$  fold change in RNA-Seq expression levels of putative target genes of ATF6, ATF4, and XBP1 in 6- vs. 30-month-old mice muscle. Central lines represent median values and box limits represent upper and lower quartiles. Whiskers represent  $1.5 \times$  the interquartile range above and below the upper and lower quartiles, respectively. **h** mRNA levels of *Atf6*, *Perk*, *Xbp1*, and their well-characterized target genes in 6- and 30-month-old mice measured by

real-time PCR ( $n = 4$ ). **i** *Xbp1s* mRNA levels in 18-month-old mice measured by real-time PCR ( $n = 5$ ). Mice were treated with a single intravenous injection of control AAV or AAV encoding mouse *Xbp1s* under the control of MHCK7 promoter 3 months before the measurements. **j–l** Rotarod performance (**j**), running endurance (**k**), and grip strength (**l**) of 9- and 18-month-old mice. Mice were treated with a single intravenous injection of control AAV or AAV encoding mouse *Xbp1s* under the control of MHCK7 promoter 3 months before the measurements ( $n = 5$  for control AAV-transduced 9-month-old mice,  $n = 26$  for control AAV-transduced 18-month-old mice, and  $n = 25$  for *Xbp1s* AAV-transduced 18-month-old mice). Horizontal lines indicate the median. Error bars are presented as the mean  $\pm$  SD values. \* (Adjusted-)  $p$  value  $< 0.05$  (two-tailed Sidak's multiple comparison test for (**b**) two-tailed  $t$ -test for (**f**, **i**) two-tailed Wilcoxon test for (**g**) two-tailed  $t$ -test corrected by BKJ method for multiple comparisons for (**h**) and Dunnett's multiple comparisons test for (**j–l**)).

changes of downstream transcription targets of the three major branches of UPR pathways, namely S1P-ATF6, PERK-ATF4, and IRE1-XBP1. XBP1- but not ATF4-binding genes with the top 100 highest scores in ChIP-Atlas database<sup>42</sup> showed increased expression in the aged muscle (Fig. 8g). ChIP-Seq data was not available for mouse ATF6, but genes with evolutionary conserved ATF6 binding sequence identified by Nair et al.<sup>43</sup> did not show overall differences in expression between 6 and 30-month-old mouse muscle (Fig. 8g). Real-time quantitative PCR of UPR regulators and their well-characterized transcriptional targets also showed that XBP1 pathway but not ATF4 and ATF6 pathways was activated in the aged muscle. *Xbp1*, *Xbp1s* (active

spliced form of *Xbp1*), and its target genes *Dnajb9* and *P4hb* were upregulated in the aged muscle, while none of the investigated genes involved in PERK and ATF6 pathways were significantly differentially expressed with age (Fig. 8h). To evaluate the functional consequences of XBP1 activation in the muscle, we treated 15-month-old mice with a single intravenous injection of AAV encoding mouse *Xbp1s* under the control of MHCK7 promoter<sup>44</sup> (Fig. 8i). At 3 months after transduction, mice transduced with AAV encoding *Xbp1s* showed lower exercise performance in rotarod test, running endurance test, and grip strength test compared to mice transduced with control AAV containing non-coding stuffer DNA (Figs. 8j–l). These results suggest that increased

Xbp1s expression in the aged muscle contributes to the age-related decline in exercise capacity.

## Discussion

We constructed the Mouse aging proteomic atlas (<https://aging-proteomics.info/>), which provides an integrative view of age-related changes in the proteome and transcriptome in a user-friendly manner (Supplementary Fig. 11). The inclusion of LSF in the analysis not only doubled the number of DEPs but also allowed in-depth exploration of age-related changes in core matrisome and mitochondrial membrane proteins, which were the major features of the aging proteome. In addition, the inclusion of geriatric mice enabled detailed investigation of DEPs that were not accompanied by significant concordant changes in mRNA levels. These features of the mouse aging proteomic atlas enhance its usefulness as a resource for future studies of aging.

We found that core matrisome and mitochondrial membrane proteins were enriched in DEPs and were likely to be more affected by post-transcriptional mechanisms than the other proteins in multiple tissues. Disproportionate age-related post-transcriptional dysregulation in mitochondrial membrane proteins might be explained by qualitative or quantitative alterations in the mitochondrial membrane, changes in trafficking or maintenance mechanisms of mitochondrial membrane proteins, or higher susceptibility of mitochondrial membrane proteins to changes in mitochondrial fission, fusion, or mitophagy. Mitochondrial fission process 1 (MTF1), which was decreased during aging in the heart, lung, and muscle (Fig. 3e), has been recently reported to play a role in preserving inner mitochondrial membrane integrity apart from its role in mitochondrial fission<sup>45</sup>. Constitutive knockout of MTF1 most severely affects the heart<sup>45</sup>, where the disproportionate decrease in mitochondrial membrane proteins during aging was most evident (Fig. 3d). It is also worth mentioning that lifespan-extending interventions such as calorie restriction and rapamycin treatment increase the half-life of OXPHOS proteins in the old mouse heart<sup>46</sup>. Correcting the imbalance between transcripts and proteins of age-dependent OXPHOS genes may thus also have anti-aging effects. However, although it has been shown that mitochondrial transcripts<sup>1,47,48</sup> and proteins<sup>49,50</sup> decrease with age in various species, whether age-related imbalance between transcripts and proteins of OXPHOS genes that was observed in mice in the current study also applies to human remains to be explored.

The reason why core matrisome proteins were preferentially increased during aging especially at the post-transcriptional level may be partially explained by the observation that cross-linking among core matrisome proteins increases with age<sup>51</sup>. Cross-linking of core matrisome proteins, which is likely to be at least partially associated with higher stiffness and fibrotic changes in aged tissues, can inhibit their efficient removal<sup>51</sup>. The formation of aggregate-like structures that do not involve cross-linking may similarly contribute to the age-related increase in core matrisome proteins. Although core matrisome proteins that increased during aging were significantly different in different tissues, some extracellular proteins, such as APOE, which is known to bind to and co-aggregate with amyloid- $\beta$ <sup>52,53</sup>, were increased in multiple tissues. It is thus possible that these commonly increased extracellular proteins contribute to the formation of extracellular deposition in multiple tissues. Similarly to aggregation, movement of the proteins between LSF and more soluble fractions may be contributing to the discrepancy between age-related changes in transcriptome and proteome. Considering its low protein abundance (Supplementary Fig. 1b), LSF proteome is likely to be more affected by such movement, which may partially explain why DEPs were more enriched in LSF than in WTL.

The comprehensiveness of proteomic mass spectrometry analysis is still far from complete and many important proteins, especially those expressed at low levels, were missed in this study. Another major limitation of this study is the lack of human data. Although human

aging proteome has been studied in great detail in muscle<sup>49</sup> and plasma<sup>20,54,55</sup>, it still needs to be experimentally investigated in future studies whether features of aging proteome found in this study also apply to humans of diverse genetic backgrounds. Finally, the functional relevance of age-related proteome changes remains to be elucidated. Our results suggest that an increase in UPR gene products accompanied by *Xbp1s* upregulation in the aged muscle contributes to the age-related decline in exercise capacity. This is consistent with a previous finding that XBP1 activation induces muscle wasting in cachexia<sup>56</sup>. It should be noted, however, that XBP1 can be activated upon muscle injury and that its deletion diminishes muscle regeneration<sup>57</sup>. In the aged mouse brain, basal *Xbp1s* levels are unchanged, but its induction upon ER stress is impaired, and the overexpression of *Xbp1s* rejuvenates the aged brain<sup>58</sup>. These observations, together with many other studies, indicate that XBP1 and UPR have both protective and deleterious roles in aging processes<sup>41</sup>. We previously showed that mRNA expression profiles associated with longer lifespan and aging exhibit similarity in mammals<sup>59</sup>, and thus it is possible that a significant portion of DEPs has adaptive rather than pro-aging role, and thus their implication in aging need to be carefully evaluated.

## Methods

### Mice

Male C57BL/6J mice were purchased from the Jackson Laboratory Japan and female C57BL/6J mice were provided by the Foundation for Biomedical Research and Innovation at Kobe through the National BioResource Project of the MEXT, Japan. C57BL/6J mice were maintained under specific pathogen-free conditions on a 12 h light-dark cycle. All procedures were approved by the Institutional Animal Care and Use Committee at Osaka Metropolitan University. For tissue collection, mice were first euthanized by cervical dislocation and then the abdominal hair was removed with depilatory cream Veet (Reckitt Benckiser). After 30 seconds, the cream was wiped off and abdominal skin, brain (cerebrum), gastrocnemius muscle, heart, kidney, liver, lung, and thoracic aorta were collected. Tissue samples were harvested without prior perfusion or blood collection. Collected tissues were immediately frozen in liquid nitrogen (except for the heart samples used for immunofluorescence staining). Frozen tissues of CB6F1 and BALB/c mice were obtained from NIA Aged Rodent Tissue Bank. Frozen tissues were thoroughly pulverized with mortar in liquid nitrogen and kept in  $-80^{\circ}\text{C}$  until use.

### Protein extraction, quantification, trypsin digestion, and peptide purification

WTL was prepared by rotating tissue powder in  $2 \times$  Laemmli buffer (4% SDS, 20% glycerol, 0.02% bromophenol blue, 125 mM Tris-Cl, pH 6.8) for 2 h at RT. Insoluble material was removed by centrifugation ( $17,700 \times g$ , 15 min, RT). A part of the WTL used in the experiment shown in Fig. 4 was prepared by rotating tissue powder in 8 M urea buffer (8 M urea, 150 mM NaCl, 50 mM Tris-Cl, pH 8.0) supplemented with protease inhibitor cocktail (Nacalai Tesque Cat No. 04080) for 2 h at RT. Insoluble material was removed by centrifugation ( $17,700 \times g$ , 15 min, RT). To prepare LSF, tissue powder was first rotated for 72 h at RT in 0.125, 0.25, 0.5, or 1% SDS/water solution supplemented with protease inhibitor cocktail (Nacalai Tesque Cat No. 04080). Pre-incubated tissues were pelleted by centrifugation ( $14,000 \times g$ , 15 min, RT) and the supernatant was discarded. The pelleted tissue was resuspended in water containing protease inhibitor cocktail (Nacalai Tesque Cat No. 04080) and rotated for 48 h at RT. Then the tissue was centrifuged again ( $17,700 \times g$ , 15 min, RT). The supernatant was discarded and the pelleted tissue was resuspended and rotated in  $2 \times$  Laemmli buffer for 2 h at RT. Insoluble material was removed by centrifugation ( $17,700 \times g$ , 15 min, RT) and the supernatant was used as LSF. For the investigation of age-related proteome change, LSF was



prepared by pre-incubating tissue powder in 0.125% SDS solution for heart, kidney, lung, and skin and in 0.25% SDS solution for the muscle. Protein concentration was measured using the DC protein assay kit (Bio-Rad Cat No. 500-0111). Trypsin digestion and peptide purification was performed using the S-Trap mini column (Protifi) following the manufacturer's instructions, except for a part of the samples used in the experiment shown in Fig. 4, which was processed using the SP3 (single-pot solid-phase-enhanced sample-preparation) protocol<sup>25</sup>.

### Proteomic mass spectrometry analysis using Triple TOF 5600 LC-MS/MS system

Purified trypsin-digested peptides were resuspended in 0.1% formic acid and separated using Nano-LC-Ultra 2D-plus equipped with cHiPLC Nanoflex (Eksigent) in trap-and-elute mode, with trap (200  $\mu\text{m} \times 0.5$  mm ChromXP C18-CL 3  $\mu\text{m}$  120 Å) and analytical column (75  $\mu\text{m} \times 15$  cm ChromXP C18-CL 3  $\mu\text{m}$  120 Å). Separation was carried out using a binary gradient in which 0.1% formic acid/water and 0.1% formic acid/acetonitrile were used as solvent A and B, respectively. The gradient program was as follows; 2 to 33.2% B for 250 min, 33.2 to 98% B in 2 min, 98% B for 5 min, 98 to 2% B in 0.1 min, and 2% B for 17.9 min. The flow rate was 300 nL/min. The analytical column temperature was set to 40 °C. The eluates were infused on-line to TripleTOF 5600 System with NanoSpray III source and heated interface (SCIEX) and ionized in an electrospray ionization-positive mode. Data acquisition was carried out with an information-dependent acquisition method. The acquired datasets were analyzed using ProteinPilot software v.5.0.1 (SCIEX). Proteins were considered to be identified if two or more peptides matched a single reference in the database with confidence at least 95%. Only unique peptides were used for protein quantification. The percentage of core matrisome and mitochondrial membrane protein-derived peptides within the total peptides shown in Supplementary Fig. 1e was calculated using the number of unique peptides. Unique peptides were assigned to proteins as described above and the lists of core matrisome proteins and mitochondrial membrane proteins were obtained from Matrisome DB<sup>18</sup> and MitoCarta 3.0<sup>19</sup>, respectively. The intensity of core matrisome proteins, mitochondrial membrane proteins, and other proteins was calculated as the sum of the weighted sum of the heights of the isotope series at the determined apex of the elution of corresponding peptides, after doing a 3-point moving average smoothing.

### Quantitative proteomic mass spectrometry analysis using multiplex TMT labeling and Orbitrap Fusion Lumos Tribrid mass spectrometer

Samples to be compared (Supplementary Data 1) were processed in the same batch. For each sample, 20  $\mu\text{g}$  of peptides were labeled with TMT<sup>60</sup> Isobaric Label Reagent (Thermo Fisher Scientific) according to the manufacturer's instructions. The reaction was quenched with 5% hydroxylamine for 15 min at RT. Samples were combined and cleaned up using an OASIS HLB  $\mu\text{Elution}$  Plate (Waters). Off-line high pH reverse phase fractionation was carried out on an Agilent 1200 Infinity high-performance liquid chromatography system, equipped with a Gemini C18 column (3  $\mu\text{m}$ , 110 Å, 100  $\times$  1.0 mm, Phenomenex). An UltiMate 3000 RSLC nano LC system (Dionex) equipped with an  $\mu$ -Precolumn (C18 PepMap 100, 5  $\mu\text{m}$ , 300  $\mu\text{m}$  i.d.  $\times$  5 mm, 100 Å) and an analytical column (nanoEase M/Z HSS T3 column 75  $\mu\text{m} \times$  250 mm C18, 1.8  $\mu\text{m}$ , 100 Å, Waters) was coupled directly to an Orbitrap Fusion Lumos Tribrid Mass Spectrometer (Thermo Fisher Scientific) using the Nanospray Flex ion source in positive ion mode. Sample trapping was carried out with a constant flow of 0.05% trifluoroacetic acid in water at 30  $\mu\text{L}/\text{min}$  for 6 min. Peptides were eluted from the analytical column running using solvent A (0.1% formic acid in water, 3% DMSO) with a constant flow of 0.3  $\mu\text{L}/\text{min}$  in combination with an increasing percentage of solvent B (0.1% formic acid in acetonitrile, 3% DMSO).

The gradient was as follows: from 2% to 8% in 4 min, from 8% to 26% in 104 min, from 28–38% in 4 min, and finally from 38%–80% in 4 min followed by re-equilibration back to 2% B in 4 min. The peptides were introduced into the Fusion Lumos via a Pico-Tip Emitter 360  $\mu\text{m}$  OD  $\times$  20  $\mu\text{m}$  ID; 10  $\mu\text{m}$  tip (CoAnn Technologies) and an applied spray voltage of 2.2 kV. The capillary temperature was set at 275 °C. Full mass scan was acquired with mass range 375–1500  $m/z$  in profile mode in the Orbitrap with resolution of 120,000. The filling time was set at a maximum of 50 ms with a limitation of  $4 \times 10^5$  ions. Data-dependent acquisition (DDA) was performed with the resolution of the Orbitrap set to 30,000, with a fill time of 94 ms and a limitation of  $1 \times 10^5$  ions. A normalized collision energy of 34 was applied. MS<sup>2</sup> data was acquired in profile mode. The acquired RAW files were searched against a *Mus musculus* (UP000000589, June 2020, 63854 entries) Uniprot database containing common contaminants and reversed sequences using IsobarQuant<sup>61</sup> and Mascot (v2.2.07). The following modifications were included into the search parameters: Carbamidomethyl (C) and TMT16 (K) as fixed modifications, Acetyl (Protein N-term), Oxidation (M) and TMT10 or TMT16 (N-term) as variable modifications. For the full scan (MS1) a mass error tolerance of 10 ppm and for MS/MS (MS2) spectra of 0.02 Da were used. Further parameters were trypsin as a protease with an allowance of a maximum of two missed cleavages, a minimum peptide length of seven amino acids, and at least two unique peptides required for protein identification. The false discovery rate on peptide and protein levels was set to 0.01. The mass spectrometry data have been deposited to the ProteomeXchange Consortium via the PRIDE partner repository<sup>62</sup> with the dataset identifier [PXD040722](https://doi.org/10.26434/chemrxiv-2024-pxd04). The raw output files of IsobarQuant were processed using the R programming language. Contaminants were filtered out. Log2 transformed raw TMT reporter ion intensities were normalized using vsn<sup>63</sup>. Missing value imputation was not conducted. Proteins were tested for differential expression using the limma package. The replicate information was added as a factor in the design matrix given as an argument to the 'lmFit' function of limma. The percentage of core matrisome and mitochondrial membrane protein-derived peptides within the total peptides shown in Fig. 1c was calculated using the number of unique peptides. Unique peptides were assigned to proteins using Mascot (v2.2.07) as described above and the lists of core matrisome proteins and mitochondrial membrane proteins were obtained from Matrisome DB<sup>18</sup> and MitoCarta 3.0<sup>19</sup>, respectively. The intensity of core matrisome proteins, mitochondrial membrane proteins, and other proteins was calculated as the sum of TMT intensities of corresponding peptides.

### Western blotting

Protein in WTL was reduced by adding 10% 2-mercaptoethanol and denatured at 95 °C for 5 min. The reduced denatured protein was separated by SDS-PAGE and transferred onto PVDF membrane. After the transfer, the membrane was first stained with Ponceau S solution and scanned. Membrane was then destained with TBST, blocked with 2% milk in TBST for 30 min, and probed with the following primary antibodies: anti-APOE (ABclonal Biotechnology Cat No. A16344), anti-Endostatin (R&D systems Cat No. AF570), anti-HSPA5 (Proteintech Cat No. 11587-1-AP), anti-Vinculin (Sigma Cat No. V9131). All primary antibodies were used at 1:1000 dilutions. After washing 4 times with TBST, the membrane was incubated with 1:1,000 diluted HRP-conjugated secondary antibody (Cytiva) for 30 min at RT, washed 4 times with TBST, and visualized with Amersham ECL reagents (Cytiva). Uncropped gel images are provided in Supplementary Fig. 12.

### Immunofluorescence staining

The heart was collected from euthanized mice, fixed with 4% paraformaldehyde for 24 h at RT, and dehydrated in 70% ethanol for

24 h at RT. Then the heart was excised, embedded in paraffin, and cut into sections of 4  $\mu\text{m}$  thickness. Sections were deparaffinized with xylene and rehydrated in graded alcohols. Antigen retrieval was performed by autoclave (121  $^{\circ}\text{C}$ , 20 min) in Target Retrieval Solution (Dako Cat No. S1699). After antigen retrieval, the sections were blocked with Power Block Universal Blocking reagent (Biogenex, Cat No. HK085-5KE) for 10 min at RT, washed twice with PBS, and incubated in PBS containing 1:200 diluted antibodies against APOE (Cell Signaling Technology Cat No. 68587) and laminin  $\alpha 2$  (Santa Cruz Biotechnology Cat No. sc-59854) at 4  $^{\circ}\text{C}$  for overnight. Sections were then washed 3 times with PBS, incubated in PBS containing 1:500 diluted secondary antibodies conjugated with Alexa Fluor 488 or 594 (Thermo Fisher Scientific Cat No. A11037 and A11006) for 1 h at RT, washed 2 times with PBS, stained with DAPI, and mounted in ProLong Gold mounting medium (Thermo Fisher Scientific Cat No. P36930). Imaging was performed using fluorescence microscope BZ-8000 (Keyence) and LSM800 confocal microscope (Zeiss).

### Gene and pathway enrichment analysis

GO and pathway enrichment analysis was performed using GOrilla<sup>64</sup> and Enrichr<sup>65</sup>. GO term enrichment in DEPs was analyzed by GOrilla using the total protein detected in the corresponding tissue as the background. Results of GO term enrichment analyzes were clustered based on Louvain graph community method using simplify Enrichment R package<sup>66</sup>.

### RNA isolation, reverse transcription, and real-time PCR

Tissue powder was homogenized with a pestle in RNAiso Plus (Takara Cat No. 9108), and the total RNA was extracted following the manufacturer's instructions. Reverse transcription was carried out using PrimeScript RT reagent kit (Takara Cat No. RR036A). Real-time PCR was performed using TB Green Premix Ex Taq II (Takara Cat No. RR820B) with StepOne-Plus real-time PCR system (Applied Biosystems). The primers used in this study are given in Supplementary Data 13.

### Bulk RNA-Seq analysis

RNA samples were sent to Genome-Lead Inc. for RNA-Seq. The sequencing libraries were prepared using the Illumina Stranded mRNA Prep kit (Illumina Cat No. 20040534) following the manufacturer's instructions. Paired-end sequencing (2  $\times$  151 bp) was performed on a NovaSeq 6000 system (Illumina) using the NovaSeq 6000 S2 Reagent Kit v.1.5 (300 cycles) (Illumina). Samples to be compared, i.e., samples from the same tissue, were processed in the same batch. Demultiplexing, quality control, and adapter trimming were performed using bcl-convert v 3.10. Poor quality reads were removed using Fastp (v.0.22.0). Clean reads were mapped to the mouse reference genome (GRCm39) using STAR (v.2.7.10b)<sup>67</sup>. The read counts were calculated using RSEM (v.1.3.1)<sup>68</sup>. Normalization and differential expression analysis were carried out using DESeq2 (v.1.36.0)<sup>69</sup>. Raw and processed data have been deposited in the Gene Expression Omnibus database under the accession number [GSE225576](https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE225576). Non-coding genes were not considered in the downstream analyses.

### Re-analysis of public snRNA-Seq data

Public single-nucleus RNA-Seq data of normal adult male C57BL/6 mouse heart<sup>30</sup>, kidney<sup>31</sup>, liver<sup>32</sup>, lung<sup>33</sup>, and muscle<sup>34</sup> were processed using Cell Ranger v.6.0.2 (10X Genomics) and aligned to the mm10 reference genome. Datasets were quality controlled, filtered, integrated, and clustered using Seurat v.4.3.0<sup>70</sup>. Cells with < 500 genes and > 10% mitochondrial-derived UMI counts were removed from the gene expression matrix. The top 7.5% of cells based on the doublet score were excluded using DoubletFinder v.2.0.3<sup>71</sup>. Expression data were log-normalized and scaled using the

ScaleData function. First 30 principal components and unsupervised shared nearest neighbor algorithm were used for the clusterization (resolution = 0.7 for heart, 3.7 for kidney, 0.1 for liver, 2.1 for lung, and 1.5 for muscle). Clusters were visualized with UMAP<sup>72</sup> and classified based on the expression of cell-type marker genes (Supplementary Data 14). The above analyses were conducted in the R environment v.4.1.0 and v.4.1.2.

### AAV production

AAV1 encoding XBP1s under MHCK7 promoter and the control empty AAV vector were purchased from VectorBuilder. AAV was produced by co-transfecting HEK293T cells (ATCC Cat No. CRL-3216) with pAAV, pRC1, and pHelper (Takara Cat No. 6672) plasmid vectors (4.2, 4.2, and 5.6  $\mu\text{g}$ , respectively, per 10-cm dish) using PEI Max reagent. AAV was purified from the cells and culture medium collected at 48, 72, 96, and 120 h post-transfection using iodixanol density gradient centrifugation method as described previously<sup>73</sup>. Mice were treated with a single intravenous injection of AAV at a dose of  $1 \times 10^{13}$  vg/mouse.

### Exercise performance tests

Rotarod performance was tested using a rotarod treadmill MK-630B (Muromachi Kikai). The mice were placed on the rod, which was gradually accelerated from 4 to 40 rpm over 300 s, and the latency to fall was recorded. Test was performed 2 times with 15 min intervals and the maximum score was used for the following analysis. Running test was conducted using a linear treadmill MK-690S (Muromachi Kikai). Mice were first acclimatized to treadmill running by performing running session at 7 m/min for 5 min. On the following day, mice were run on a treadmill under the following conditions: the starting speed was 9 m/min; the speed was increased 2 m/min every 3 min until it reached 17 m/min; thereafter, the speed continued to increase by 1 m/min every 3 min until the mice were unable to remain on the treadmill despite an electrical shock. The forelimb gripping strength was assessed 3 times with at least 10 min intervals using a grip strength meter MK-380Si (Muromachi Kikai) and the average score was used for the analysis.

### Statistical analysis

All experiments were performed with a minimum of 4 biological replicates. All statistical tests are two-sided. Details of statistical analyzes are indicated in figure legends and exact *p*- and *q*-values are provided in Source Data.

### Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

### Data availability

Source data underlying Figs. 3a-d, 5d,e, 8b,f-i, Supplementary Figs. 1f,g, 2a,b, 3a,b, and 7c are provided in Supplementary Data 2-12 and Source Data. Mass spectrometry data have been deposited in the ProteomeXchange with identifier [PXD040722](https://www.ebi.ac.uk/psd/entry/PXD040722). RNA-Seq data have been deposited in the Gene Expression Omnibus database under the accession numbers [GSE225576](https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE225576). Source data are provided with this paper.

### References

- Schaum, N. et al. Ageing hallmarks exhibit organ-specific temporal signatures. *Nature* **583**, 596–602 (2020).
- TabulaMurisConsortium. A single-cell transcriptomic atlas characterizes ageing tissues in the mouse. *Nature* **583**, 590–595 (2020).
- López-Otín, C., Blasco, M. A., Partridge, L., Serrano, M. & Kroemer, G. Hallmarks of aging: an expanding universe. *Cell* **186**, 243–278 (2023).

4. Angelidis, I. et al. An atlas of the aging lung mapped by single cell transcriptomics and deep tissue proteomics. *Nat. Commun.* **10**, 963 (2019).
5. Kluever, V. et al. Protein lifetimes in aged brains reveal a proteostatic adaptation linking physiological aging to neurodegeneration. *Sci. Adv.* **8**, eabn4437 (2022).
6. Takemon, Y. et al. Proteomic and transcriptomic profiling reveal different aspects of aging in the kidney. *Elife* **10**, <https://doi.org/10.7554/eLife.62585> (2021).
7. Williams, E. G. et al. Multiomic profiling of the liver across diets and age in a diverse mouse population. *Cell Syst.* **13**, 43–57.e46 (2022).
8. Keele, G. R. et al. Global and tissue-specific aging effects on murine proteomes. *Cell Rep.* **42**, 112715 (2023).
9. Lennon, R. et al. Global analysis reveals the complexity of the human glomerular extracellular matrix. *J. Am. Soc. Nephrol.* **25**, 939–951 (2014).
10. Schiller, H. B. et al. Time- and compartment-resolved proteome profiling of the extracellular niche in lung injury and repair. *Mol. Syst. Biol.* **11**, 819 (2015).
11. Naba, A. et al. Characterization of the extracellular matrix of normal and diseased tissues using proteomics. *J. Proteome Res.* **16**, 3083–3091 (2017).
12. Elinger, D., Gabashvili, A. & Levin, Y. Suspension trapping (S-Trap) is compatible with typical protein extraction buffers and detergents for bottom-up proteomics. *J. Proteome Res.* **18**, 1441–1445 (2019).
13. Didangelos, A. et al. Proteomics characterization of extracellular space components in the human aorta. *Mol. Cell Proteom.* **9**, 2048–2062 (2010).
14. Henning, N. F., LeDuc, R. D., Even, K. A. & Laronda, M. M. Proteomic analyses of decellularized porcine ovaries identified new matrix proteins and spatial differences across and within ovarian compartments. *Sci. Rep.* **9**, 20001 (2019).
15. Thygesen, C., Metaxas, A., Larsen, M. R. & Finsen, B. Age-dependent changes in the sarkosyl-insoluble proteome of appsw/ps1 $\Delta$ e9 transgenic mice implicate dysfunctional mitochondria in the pathogenesis of Alzheimer's disease. *J. Alzheimers Dis.* **64**, 1247–1259 (2018).
16. Kelmer Sacramento, E. et al. Reduced proteasome activity in the aging brain results in ribosome stoichiometry loss and aggregation. *Mol. Syst. Biol.* **16**, e9596 (2020).
17. Molzahn, C. et al. Shift of the insoluble content of the proteome in the aging mouse brain. *Proc. Natl Acad. Sci. USA* **120**, e2310057120 (2023).
18. Shao, X. et al. MatrisomeDB 2.0: 2023 updates to the ECM-protein knowledge database. *Nucleic Acids Res.* **51**, D1519–D1530 (2023).
19. Rath, S. et al. MitoCarta3.0: an updated mitochondrial proteome now with sub-organelle localization and pathway annotations. *Nucleic Acids Res.* **49**, D1541–D1547 (2021).
20. Lehallier, B. et al. Undulating changes in human plasma proteome profiles across the lifespan. *Nat. Med.* **25**, 1843–1850 (2019).
21. Yang, Y. R. et al. Plasma proteomic profiling of young and old mice reveals cadherin-13 prevents age-related bone loss. *Aging (Albany NY)* **12**, 8652–8668 (2020).
22. Vidal, R. et al. Transcriptional heterogeneity of fibroblasts is a hallmark of the aging heart. *JCI Insight* **4**, <https://doi.org/10.1172/jci.insight.131092> (2019).
23. Wang, S., Dong, D., Li, X. & Wang, Z. pan-tissue transcriptome analysis reveals sex-dimorphic human aging. Preprint at *bioRxiv*, <https://doi.org/10.1101/2023.05.26.542373> (2023).
24. Park, S. K. et al. Gene expression profiling of aging in multiple mouse strains: identification of aging biomarkers and impact of dietary antioxidants. *Aging Cell* **8**, 484–495 (2009).
25. Hughes, C. S. et al. Single-pot, solid-phase-enhanced sample preparation for proteomics experiments. *Nat. Protoc.* **14**, 68–85 (2019).
26. Lin, M. K. et al. HTRA1, an age-related macular degeneration protease, processes extracellular matrix proteins EFEMP1 and TSP1. *Aging Cell* **17**, e12710 (2018).
27. Bordi, M. et al. A gene toolbox for monitoring autophagy transcription. *Cell Death Dis.* **12**, 1044 (2021).
28. Aman, Y. et al. Autophagy in healthy aging and disease. *Nat. Aging* **1**, 634–650 (2021).
29. Li, P. et al. Autophagy and aging: roles in skeletal muscle, eye, brain and hepatic tissue. *Front Cell Dev. Biol.* **9**, 752962 (2021).
30. Li, W. et al. Single-cell RNA-seq of heart reveals intercellular communication drivers of myocardial fibrosis in diabetic cardiomyopathy. *Elife* **12**, <https://doi.org/10.7554/eLife.80479> (2023).
31. Lu, Y. A. et al. Single-nucleus RNA sequencing identifies new classes of proximal tubular epithelial cells in kidney fibrosis. *J. Am. Soc. Nephrol.* **32**, 2501–2516 (2021).
32. Nault, R., Fader, K. A., Bhattacharya, S. & Zacharewski, T. R. Single-nuclei RNA sequencing assessment of the hepatic effects of 2,3,7,8-tetrachlorodibenzo-p-dioxin. *Cell Mol. Gastroenterol. Hepatol.* **11**, 147–159 (2021).
33. Koenitzer, J. R., Wu, H., Atkinson, J. J., Brody, S. L. & Humphreys, B. D. Single-nucleus RNA-sequencing profiling of mouse lung. reduced dissociation bias and improved rare cell-type detection compared with single-cell RNA sequencing. *Am. J. Respir. Cell Mol. Biol.* **63**, 739–747 (2020).
34. Petrany, M. J. et al. Single-nucleus RNA-seq identifies transcriptional heterogeneity in multinucleated skeletal myofibers. *Nat. Commun.* **11**, 6374 (2020).
35. Abondio, P. et al. The genetic variability of apoe in different human populations and its implications for longevity. *Genes (Basel)* **10**, <https://doi.org/10.3390/genes10030222> (2019).
36. Shinohara, M. et al. APOE2 is associated with longevity independent of Alzheimer's disease. *Elife* **9**, <https://doi.org/10.7554/eLife.62199> (2020).
37. Santinha, D. et al. Remodeling of the cardiac extracellular matrix proteome during chronological and pathological aging. *Mol. Cell Proteom.* **23**, 100706 (2024).
38. Tsiolaki, P. L., Katsafana, A. D., Baltoumas, F. A., Louros, N. N. & Iconomidou, V. A. Hidden aggregation hot-spots on human apolipoprotein e: a structural study. *Int. J. Mol. Sci.* **20**, <https://doi.org/10.3390/ijms20092274> (2019).
39. O'Reilly, M. S. et al. Endostatin: an endogenous inhibitor of angiogenesis and tumor growth. *Cell* **88**, 277–285 (1997).
40. Shen, J., Chen, X., Hendershot, L. & Prywes, R. ER stress regulation of ATF6 localization by dissociation of BiP/GRP78 binding and unmasking of Golgi localization signals. *Dev. Cell* **3**, 99–111 (2002).
41. Martínez, G., Duran-Aniotz, C., Cabral-Miranda, F., Vivar, J. P. & Hetz, C. Endoplasmic reticulum proteostasis impairment in aging. *Aging Cell* **16**, 615–623 (2017).
42. Oki, S. et al. ChIP-Atlas: a data-mining suite powered by full integration of public ChIP-seq data. *EMBO Rep* **19**, <https://doi.org/10.15252/embr.201846255> (2018).
43. Ramdas Nair, A., Lakhiani, P., Zhang, C., Macchi, F. & Sadler, K. C. A permissive epigenetic landscape facilitates distinct transcriptional signatures of activating transcription factor 6 in the liver. *Genomics* **114**, 107–124 (2022).
44. Salva, M. Z. et al. Design of tissue-specific regulatory cassettes for high-level rAAV-mediated expression in skeletal and cardiac muscle. *Mol. Ther.* **15**, 320–329 (2007).
45. Donnarumma, E. et al. Mitochondrial Fission Process 1 controls inner membrane integrity and protects against heart failure. *Nat. Commun.* **13**, 6634 (2022).
46. Karunadharma, P. P. et al. Respiratory chain protein turnover rates in mice are highly heterogeneous but strikingly conserved across tissues, ages, and treatments. *Faseb j.* **29**, 3582–3592 (2015).



47. Yang, J. et al. Synchronized age-related gene expression changes across multiple tissues in human and the link to complex diseases. *Sci. Rep.* **5**, 15145 (2015).
48. Shavlakadze, T. et al. Age-Related Gene Expression Signature in Rats Demonstrate Early, Late, and Linear Transcriptional Changes from Multiple Tissues. *Cell Rep.* **28**, 3263–3273.e3263 (2019).
49. Ubaida-Mohien, C. et al. Discovery proteomics in aging human skeletal muscle finds change in spliceosome, immunity, proteostasis and mitochondria. *Elife* **8**, <https://doi.org/10.7554/eLife.49874> (2019).
50. Ibebunjo, C. et al. Genomic and proteomic profiling reveals reduced mitochondrial function and disruption of the neuromuscular junction driving rat sarcopenia. *Mol. Cell Biol.* **33**, 194–212 (2013).
51. Ewald, C. Y. The matrisome during aging and longevity: a systems-level approach toward defining matreotypes promoting healthy aging. *Gerontology* **66**, 266–274 (2020).
52. Wisniewski, T. & Frangione, B. Apolipoprotein E: a pathological chaperone protein in patients with cerebral and systemic amyloid. *Neurosci. Lett.* **135**, 235–238 (1992).
53. Kanekiyo, T., Xu, H. & Bu, G. ApoE and A $\beta$  in Alzheimer's disease: accidental encounters or partners? *Neuron* **81**, 740–754 (2014).
54. Ahadi, S. et al. Personal aging markers and ageotypes revealed by deep longitudinal profiling. *Nat. Med.* **26**, 83–90 (2020).
55. Johnson, A. A., Shokhirev, M. N., Wyss-Coray, T. & Lehallier, B. Systematic review and analysis of human proteomics aging studies unveils a novel proteomic aging clock and identifies key processes that change with age. *Ageing Res Rev.* **60**, 101070 (2020).
56. Bohnert, K. R. et al. The Toll-Like Receptor/MyD88/XBP1 Signaling Axis Mediates Skeletal Muscle Wasting during Cancer Cachexia. *Mol. Cell Biol.* **39**, (2019).
57. Roy, A. et al. The IRE1/XBP1 signaling axis promotes skeletal muscle regeneration through a cell non-autonomous mechanism. *Elife* **10**, <https://doi.org/10.7554/eLife.73215> (2021).
58. Cabral-Miranda, F. et al. Unfolded protein response IRE1/XBP1 signaling is required for healthy mammalian brain aging. *Embo j.* **41**, e111952 (2022).
59. Takasugi, M., Yoshida, Y., Nonaka, Y. & Ohtani, N. Gene expressions associated with longer lifespan and aging exhibit similarity in mammals. *Nucleic Acids Res.* **51**, 7205–7219 (2023).
60. Li, J. et al. TMTpro reagents: a set of isobaric labeling mass tags enables simultaneous proteome-wide measurements across 16 samples. *Nat. Methods* **17**, 399–404 (2020).
61. Franken, H. et al. Thermal proteome profiling for unbiased identification of direct and indirect drug targets using multiplexed quantitative mass spectrometry. *Nat. Protoc.* **10**, 1567–1593 (2015).
62. Perez-Riverol, Y. et al. The PRIDE database resources in 2022: a hub for mass spectrometry-based proteomics evidences. *Nucleic Acids Res.* **50**, D543–d552 (2022).
63. Huber, W., von Heydebreck, A., Sultmann, H., Poustka, A. & Vingron, M. Variance stabilization applied to microarray data calibration and to the quantification of differential expression. *Bioinformatics* **18**, S96–S104 (2002).
64. Eden, E., Navon, R., Steinfeld, I., Lipson, D. & Yakhini, Z. GOrilla: a tool for discovery and visualization of enriched GO terms in ranked gene lists. *BMC Bioinforma.* **10**, 48 (2009).
65. Chen, E. Y. et al. Enrichr: interactive and collaborative HTML5 gene list enrichment analysis tool. *BMC Bioinforma.* **14**, 128 (2013).
66. Gu, Z. & Hubschmann, D. simplifyEnrichment: a bioconductor package for clustering and visualizing functional enrichment results. *Genomics Proteom. Bioinforma.* **21**, 190–202 (2023).
67. Dobin, A. et al. STAR: ultrafast universal RNA-seq aligner. *Bioinformatics* **29**, 15–21 (2013).
68. Li, B. & Dewey, C. N. RSEM: accurate transcript quantification from RNA-Seq data with or without a reference genome. *BMC Bioinforma.* **12**, 323 (2011).
69. Love, M. I., Huber, W. & Anders, S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biol.* **15**, 550 (2014).
70. Butler, A., Hoffman, P., Smibert, P., Papalexi, E. & Satija, R. Integrating single-cell transcriptomic data across different conditions, technologies, and species. *Nat. Biotechnol.* **36**, 411–420 (2018).
71. McGinnis, C. S., Murrow, L. M. & Gartner, Z. J. Doubletfinder: doublet detection in single-cell RNA sequencing data using artificial nearest neighbors. *Cell Syst.* **8**, 329–337.e324 (2019).
72. Becht, E. et al. Dimensionality reduction for visualizing single-cell data using UMAP. *Nat. Biotechnol.* <https://doi.org/10.1038/nbt.4314> (2018).
73. Challis, R. C. et al. Systemic AAV vectors for widespread and targeted gene delivery in rodents. *Nat. Protoc.* **14**, 379–414 (2019).

## Acknowledgements

We thank the Research Support Platform of Osaka Metropolitan University Graduate School of Medicine for supporting chemiluminescence detection of Western blots (Fusion Solo S7 image system, Vilber Lourmat), immunofluorescence microscopy (BZ-X800, Keyence), confocal microscopy (LSM800, Zeiss), and microplate reader analysis (ARVOx, PerkinElmer Life Science); Genome-Lead Inc. for performing bulk RNA-Seq analysis; the Human Genome Center of the University of Tokyo for supporting snRNA-Seq data analysis (SHIR-OKANE high-performance computing system); Dr. Hozumi Motohashi for advice on mitochondrial protein analysis; and the members in the Department of Pathophysiology of Osaka Metropolitan University Graduate School of Medicine for technical assistance and helpful discussions. This work was supported by grants to A.S. and V.G. from US National Institute of Health, to N.O. from Takeda Science Foundation (Visionary Hop), Japan Society for the Promotion of Science (JSPS-KAKENHI, Grant Number 22H03540), and AMED (Grant Numbers JP24gm4010026h0001, JP23ama221127s0101, and JP23ck0106793h0001), and to M.T. from JST, PRESTO (Grant Number JPMJPR2384), Japan Society for the Promotion of Science (JSPS-KAKENHI, Grant Number 23K10963), MSD Life Science Foundation and Public Interest Incorporated Foundation, Astellas Foundation for Research on Metabolic Disorders (Grant Number 2021A3097), Senri Life Science Foundation (Grant Number 82142-00027), Mizutani Foundation for Glycoscience (Grant Number 230001), Nakajima Foundation, Suzuken Memorial Foundation, Nakatomi Foundation, and Chugai Foundation for Innovative Drug Discovery Science (2022-II-08).

## Author contributions

M.T. conceived and designed the study and performed sample collection and most of the data analysis. K.T. and M.T. performed protein extraction and S-Trap processing. Y.N. performed t-SNE analysis and snRNA-Seq data analysis. Y.Y. preprocessed bulk-RNA-Seq data and performed fluorescence microscopy. F.S. processed and analyzed the proteomic data obtained by Orbitrap Fusion Lumos Tribrid mass spectrometer. J.J.S. performed TMT-labeling and mass spectrometry analysis using Orbitrap Fusion Lumos Tribrid mass spectrometer. Y.N. performed t-SNE analysis. J.S. and S.I. performed mass spectrometry analysis using the Triple TOF 5600 system. S.A.B. supported sample collection. K.T., Y.Y., J.A., G.T., A.S., and V.G. supported preliminary experiments. N.O. supervised the project. M.T. wrote the manuscript with input from all authors.

## Competing interests

The authors declare no competing interests.



## Additional information

**Supplementary information** The online version contains supplementary material available at <https://doi.org/10.1038/s41467-024-52845-x>.

**Correspondence** and requests for materials should be addressed to Masaki Takasugi or Naoko Ohtani.

**Peer review information** *Nature Communications* thanks Nathan Basisty, and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. A peer review file is available.

**Reprints and permissions information** is available at <http://www.nature.com/reprints>

**Publisher's note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

**Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2024